

RMUTT HPE Aruba Networking campus associate

Aruba Campus Network Training • 26 ส.ค. 2569

Agenda (1/3)

Overview Aruba Solution

Network devices & roles

Campus architecture

CX / Gateway / AP portfolio

Agenda (2/3)

Module 1 : Aruba Switch

Switch series, stacking, PoE

Lab 1: Device management

Lab 2: LAG & VLAN

Lab 3: Layer 3 Routing

Knowledge check?

Agenda (3/3)

Module 2 : Controller + AP

Controller & AP onboarding

WLAN: PSK & MAC auth

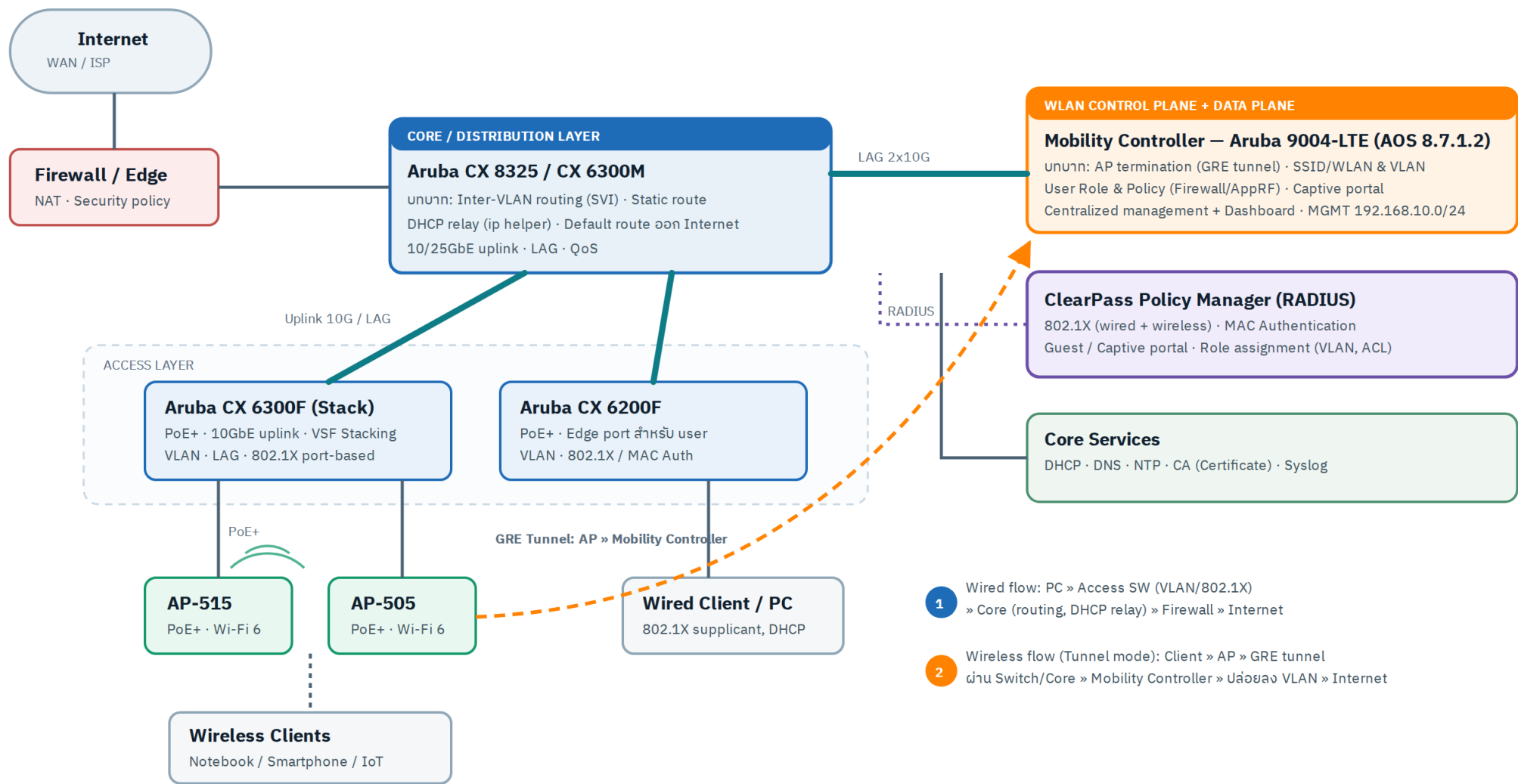
WLAN: 802.1X

WLAN: Captive portal (Guest)

Role & Policy · Q & A

Aruba Campus Architecture — Overview

Switch + Controller + AP · Device roles & traffic flow — RMUTT / Commserv Siam · 26 ส.ค. 2569



Network Devices

Networking devices: switch

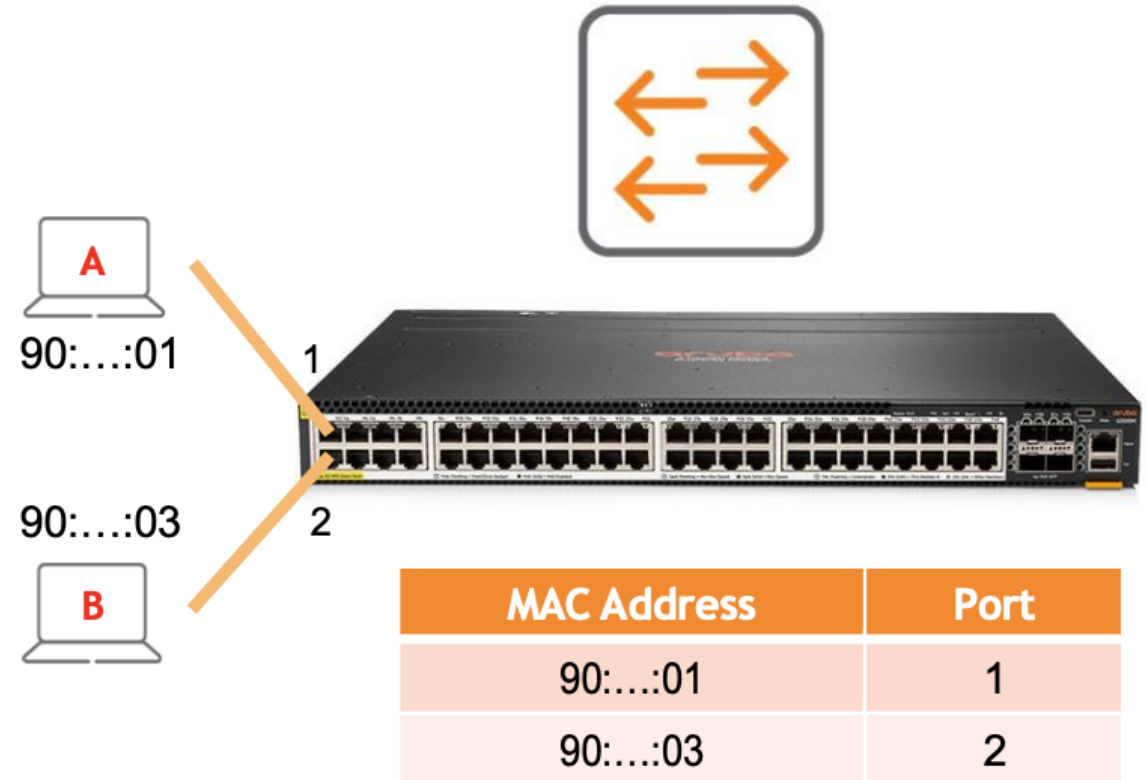
L2 device: Forwards Ethernet frames based on destination MAC addresses

Multiple ports connect endpoints

- PCs, printers, cameras, and more
- Commonly 8, 24, 48, or more physical ports
- Transparent to endpoints

Special L2 protocols

- Performance, reliability, security
- STP, LLDP, 802.1Q



Networking devices: router

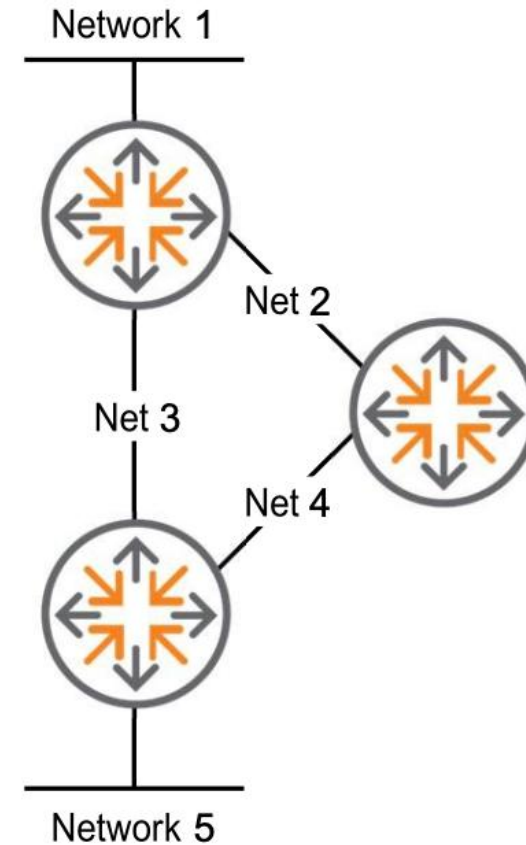
Layer 3 device: Forwards packets based on destination IP address

Functionality and features

- Connect separate networks into an inter-network
- Offer WAN connectivity to networks

L3 routing protocols

- Learn all possible paths, choose a best path
- RIP, OSPF, BGP



Networking devices: multi-layer switch

Performs both L2 switching and L3 routing

L3 Routing

- Internal routing functionality
- Runs RIP, OSPF, BGP

L2 Switching

- Multiple ports connect endpoints
- Uses STP, LLDP, VLANs



Networking devices: access point

Enables wireless users to access wired resources and roam about

Functionality

- Bridges wireless devices and wired networks.
- Transform Ethernet into Wi-Fi frames and back



Varieties of APs

- Internal or external antennas
- Single or Dual Ethernet Ports
- Indoor or outdoor



Networking devices: firewall

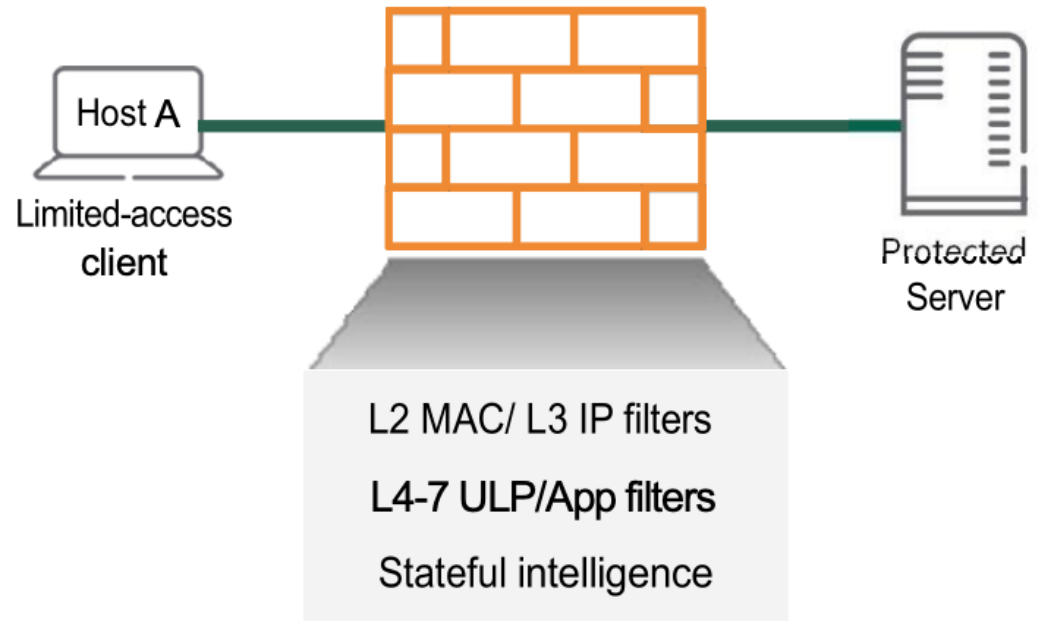
L2-7 filtering and security

Functionality

- Block unauthorized/inappropriate access
- Permit authorized/appropriate access

Types

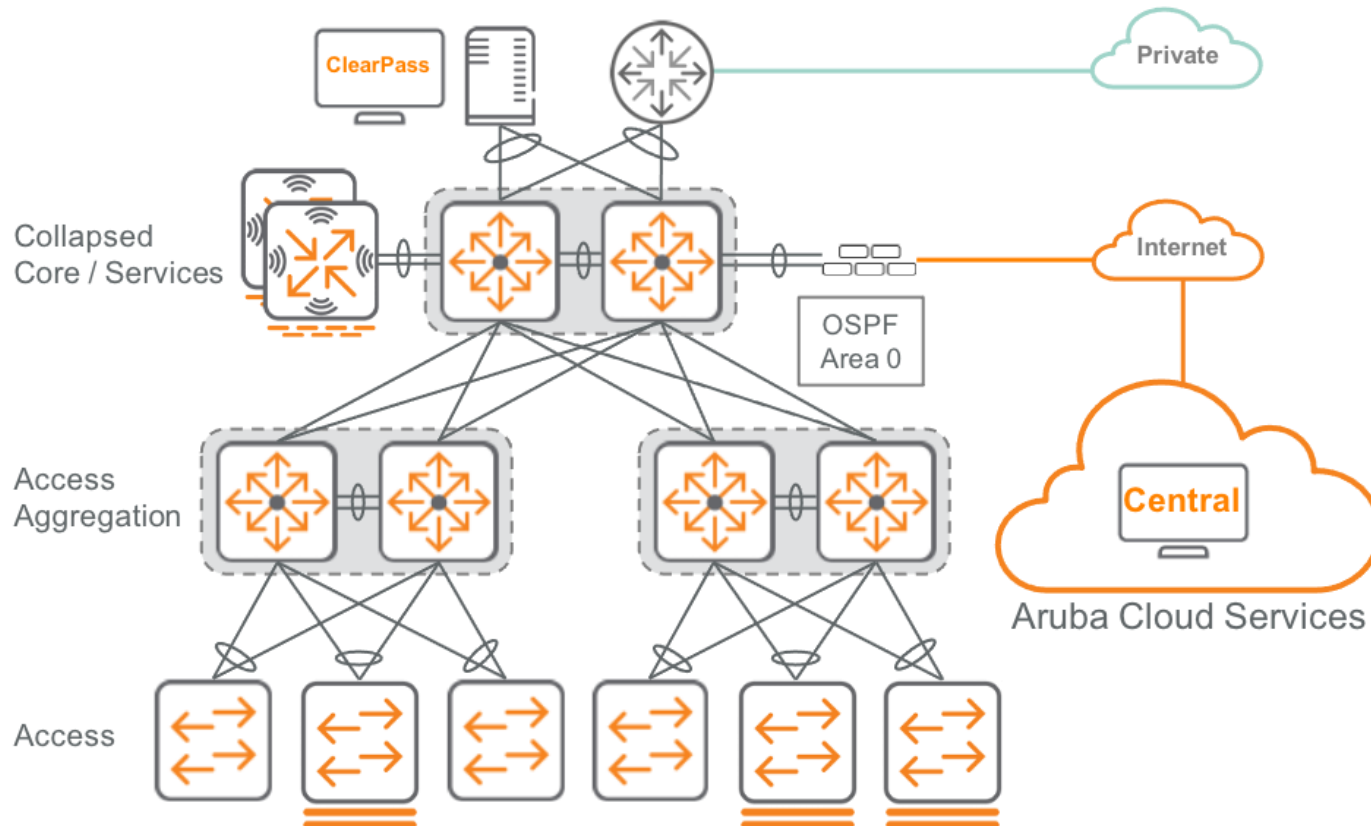
- Stateless vs Stateful
- Functionality built into switches, gateways



Introduction switch

CX switching solutions for all networks

For branch, campus, mid and large enterprises, SMB, and data center networks



Data Center

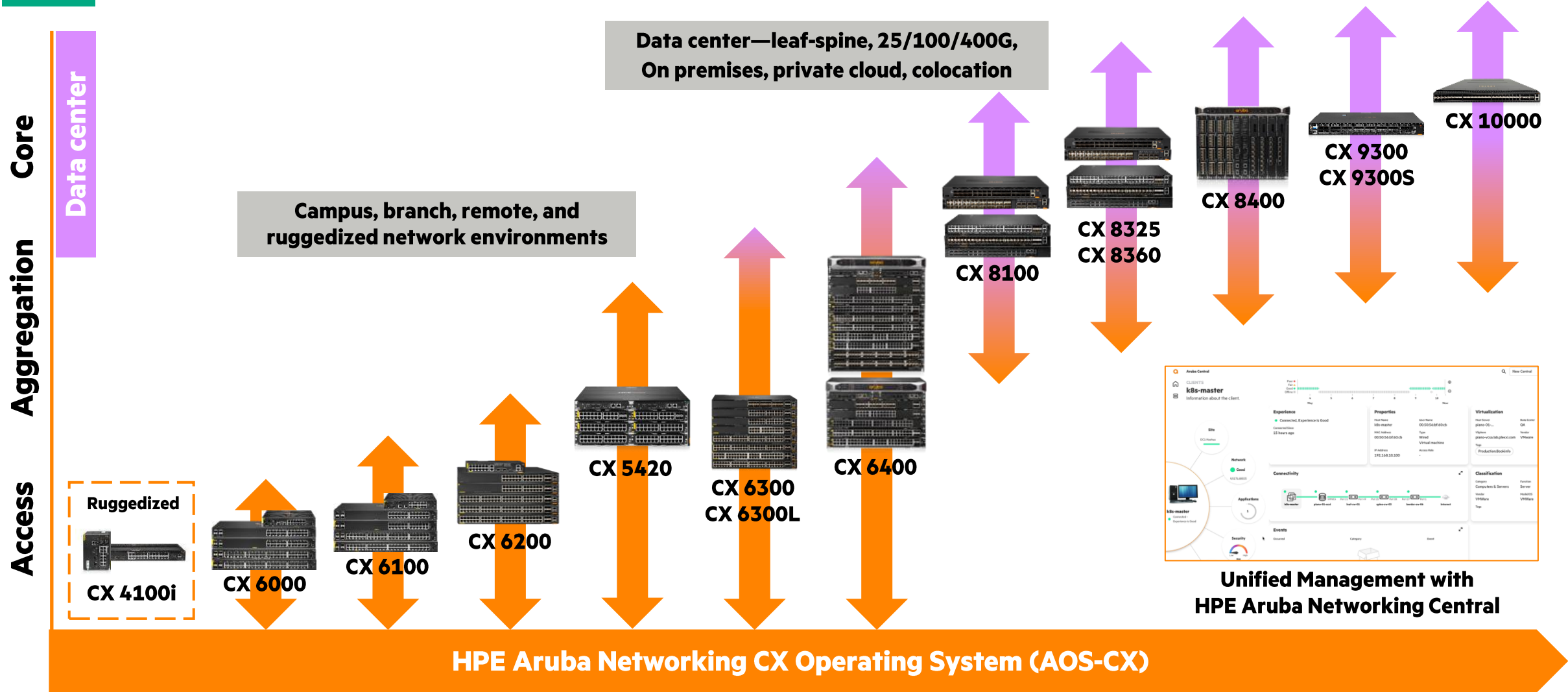
Enterprise Core

Aggregation and Distribution

Campus and Branch Access

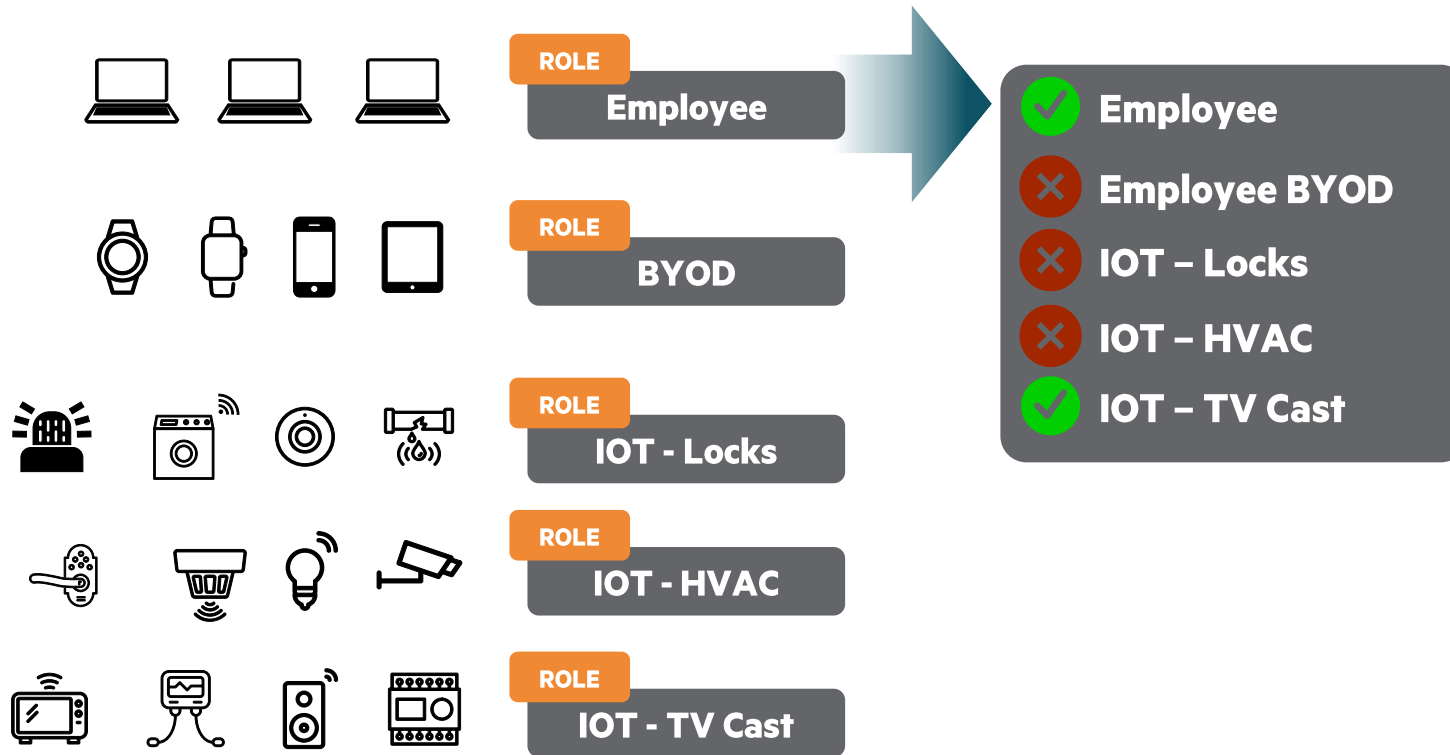
Ruggedized

HPE Aruba Networking CX switch portfolio



One common operating model from campus edge to DC

HPE Aruba Networking User Roles



DYNAMIC SEGMENTATION

Software defined approach eliminates VLAN sprawl and simplifies policy implementation

Delivers wired, wireless, and SD-WAN micro-segmentation needed for securing end-user and IoT devices

AP platforms for comprehensive campus and branch connectivity

Wi-Fi 7

Wi-Fi 6E

Wi-Fi 6

AOS-10
BASED



Mid-range Wi-Fi 7 730 Series (AP-734/735)

Tri-band support
2x2:2SS (8RU), 9.4Gbps,
2x 5GE, 2xUSB, embedded GPS &
barometer, 2 x IoT radios
Ultra tri-band filtering

AOS-10
BASED



Flagship Wi-Fi 7 750 Series (AP-734/735)

Tri-band support
4x4:4SS (8RU), 18.8Gbps,
2x 10GE, 2xUSB, embedded GPS &
barometer, 2 x IoT radios
Ultra tri-band filtering



Entry level Wi-Fi 6E 610 Series (AP-615)

Dual-radio/tri-band
2x2:2SS (8RU), 3.6Gbps,
1x 2.5GE, USB, embedded GPS



Mid-range Wi-Fi 6E 630 Series (AP-634/635)

Tri-band support
2x2:2SS (8RU), 3.9Gbps,
2x 2.5GE, USB, embedded GPS, UTB



Flagship Wi-Fi 6E 650 Series (AP-654/655)

Tri-band support
4x4:4SS (8RU), 7.8Gbps,
2x 5.0 GE, USB, embedded GPS, UTB



Entry level Wi-Fi 6 500 Series (AP-503)

Dual-radio 2x2:2SS, 1.5Gbps, 1x
1GE, USB



Entry level Wi-Fi 6 500 Series (AP-505/504)

Dual-radio 2x2:2SS (8RU), 1.5Gbps, 1x
1GE, USB, DC, BLE/Zigbee, IPM, FTM



Mid-range Wi-Fi 6 510 Series (AP-515/514)

Dual-radio 2x2:2SS (2.4GHz), 4x4:4SS (5GHz),
2.7Gbps, 1x2.5GE + 1xGE, USB, DC, BLE/Zigbee,
IPM, FTM



High-performance Wi-Fi 6 530 Series (AP-535/534)

Dual-radio 4x4:4SS, 3.98Gbps, 2x 5GE, USB,
DC, BLE/Zigbee, IPM, FTM



Flagship Wi-Fi 6 550 Series (AP-555/554)

Dual-radio 4x4:4SS (2.4GHz) 4x4:4SS x2 or
8x8:8SS (5GHz), 5.4Gbps, 2x 5GE, USB, DC,
BLE/Zigbee, IPM, FTM

< Low-density to high-density, demanding environments >

Always-on, always available enterprise networking

Resilient, scalable architecture that eliminates downtime for business continuity

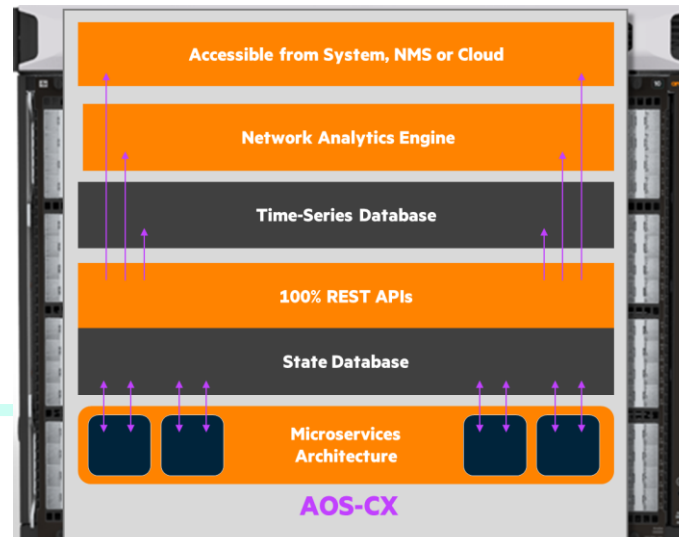


Resilient networks are designed to remain always-on, ensuring uptime for **business continuity** when every moment matters.

- Management and operations
- Physical platform redundancy
- Physical design redundancy
- Logical design redundancy
- Disaster recovery
- Software upgrades

Resilient by design

AOS-CX is a modern, cloud-native, and modular operating system built to be database-centric with independent micro-services and state synchronization.



VSX Live Upgrades

Hot Patching

VSF ISSU

Built-in analytics

Rollout validated configs

Clusters & Stacking

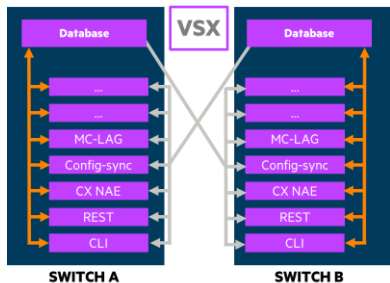
Hot-swappable parts

Always-on PoE

CX creates business continuity across the network thru SW and HW

Resilient, scalable architecture by design

VSX Live Upgrades



ISSU orchestrated upgrades to eliminate maintenance windows

Rollout validated configs

Automated rollouts with NetEdit or Central Multi-Editor

VSF ISSU & hot patching

Minimize downtime during upgrades across a stack

Clustering & Stacking

Grow quickly, simplify mgmt., extend distances with long range trvrs.

Built-in analytics

Visibility and alerts for faster troubleshooting and MTTR

Always-on PoE

Continuous power for IoT and APs during switch updates

Redundant and hot-swappable parts

Power supplies, fans, management, and fabric

Virtual Switching Framework (VSF)

For resilient and scalable campus and branch access networks

Simplified Management with virtual switch stack

Grow your network quickly

Go further with stacking across campus and site using long range transceivers

Fast stack with CX Mobile App that auto-detects potential stack members for a virtualized stack with just a few taps

Configure a stack quickly within Central UI



CX 6300

10-unit CX 6300 stack

Flexibility to combine 6300M and 6300F models

Uses 10/25/50 GbE uplinks

CX 6200

8-unit CX 6200 stack

Flexibility to combine 6200M and 6200F models

Uses 10GbE uplinks

* 6300L only stacks with other 6300L models.
6200 and 6300 mixed stack not allowed

In-Service Software Upgrades (ISSU) with VSF Stacking

Rapid, interruption-less software upgrades for stacked CX 6300 switches¹

What is an ISSU?

- Software upgrades within a release, without traffic interruption. The data plane will keep forwarding traffic during entire upgrade process
- Requires conductor and standby as part of the VSF stack.
- Supports up to 10-member stack

Benefits

- Increase availability during upgrade
- Data plane is not affected and keeps forwarding traffic
- Maintain end-points and application connectivity through the upgrade process
- Includes bug/vulnerability fixes and new features

Total Downtime: 0%

Customer starts ISSU, and observes system ready validation

Standby and members update, then connect to standby database

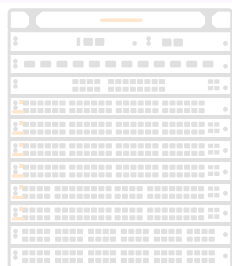
Database sync from active conductor to standby occurs

Switchover from active conductor to standby, new active takes over

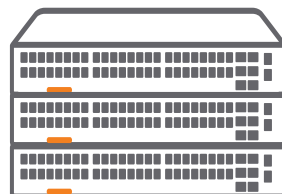
Former active updates & becomes new standby without reboot

¹ VSF ISSU not available on CX 6300L switch models

Configurations
Supported



Chassis



Stack



Standalone

CX 6200 Switch Series

Lite Layer 3 stackable access switches for branch offices and SMBs

Flexible growth with stackable gigabit connectivity with fast 1/10G uplinks

Power IoT & Wi-Fi devices with up to 1440W Class 4 to Class 6 PoE

Scalable growth with **8 member VSF stacking**

Real-time monitoring and troubleshooting with **Network Analytics Engine (NAE)**

Simple and secure access with **Dynamic Segmentation** that automates user and device policy

CX portfolio benefits

Advanced configs and programmability with AOS-CX

Single pane of glass Central across wired, wireless, WAN

Flexible management via Web GUI, CLI, NetEdit, Central

Limited lifetime warranty and options for adv.TAC support

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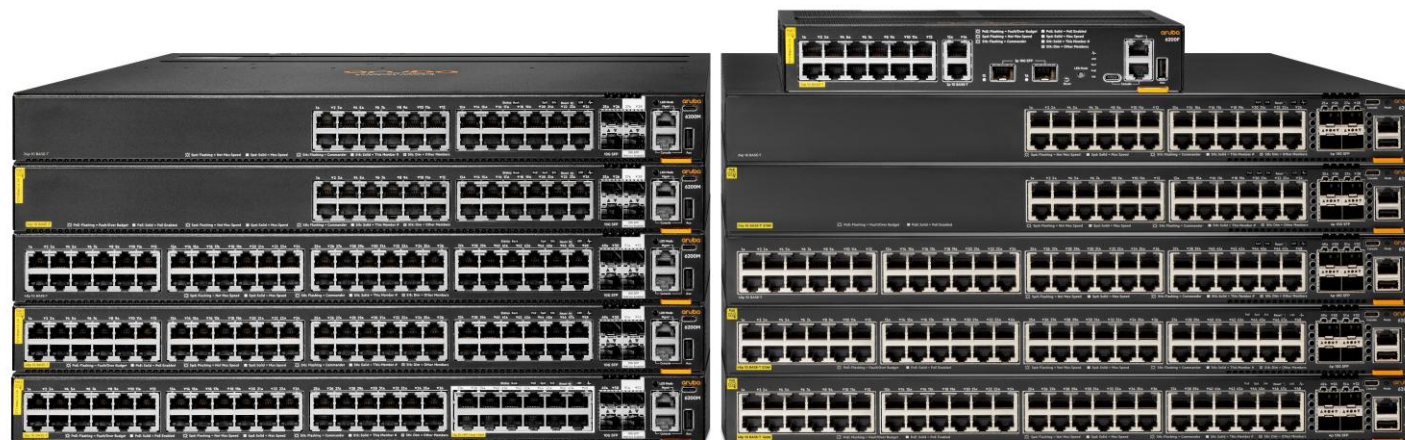
Fixed switches
(power & fans)

5

Modular switches
(power & fans)

4

Built-in 1 and 1/10 GbE uplinks



CX 6300 Switch Series

Powerful Layer 3 stackable switches ready for demanding environments

Flexible growth with stackable gigabit connectivity with fast **10/25/50GbE¹ uplinks**, high density SFP+

Built for IoT and Wi-Fi 6 with up to **2880W Class 6 and Class 8 PoE** and HPE Smart Rate **multi-gigabit** Ethernet

Real-time monitoring and troubleshooting with **Network Analytics Engine (NAE)**

Simple and secure access with **Dynamic Segmentation**; campus and branch fabric with **VXLAN and BGP EVPN**

Power-to-port airflow for cost-effective **1GbE ToR and OOBM data center** deployments

CX portfolio benefits

Advanced configs and programmability with AOS-CX

Single pane of glass Central across wired, wireless, WAN

Flexible management via Web GUI, CLI, NetEdit, Central

Limited lifetime warranty and options for adv.TAC support

4
Fixed power switches

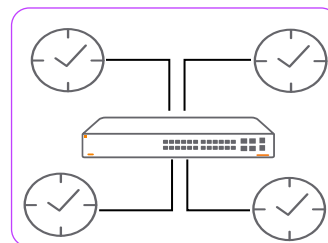
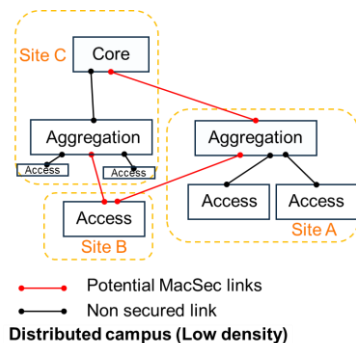
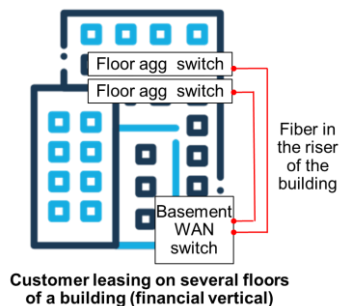
12
Modular power switches

2880W
60W and 90W PoE



10 member
VSF stacking

Bringing performance and versatility to campus networks

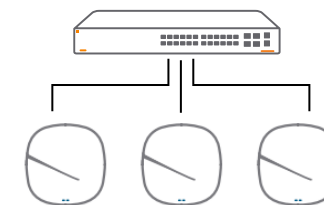


Synchronization and validation across systems

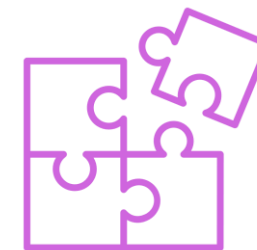


Video conferencing and multimedia consumption

Time synchronization and Audio Video Bridging support with precision time protocol (PTP) boundary clock (BC) and transparent clock (TC).

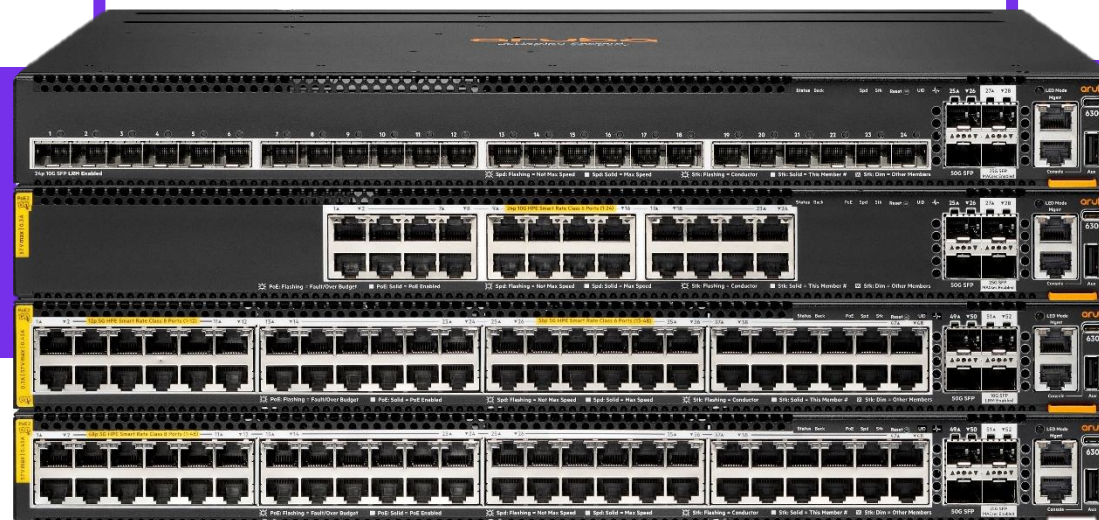


Next-gen 802.3bt enabled wireless access points



Internet of things devices and systems, like smart lighting

Stronger data link layer security with standards-based MACsec 256 encryption and authentication on uplinks and downlinks.



More performance, power, and connectivity options with up to 10G Smart Rate, Class 8 (90W) PoE, and LRM support.

HPE Aruba Networking CX Switches – End-to-end portfolio

Always-on Reliability

- HA with ISSU, HW, SW resiliency
- Proactive analytics

Switching Solution for Any Network

Strong, scalable switching portfolio for branch, campus, mid and large enterprises, SMB, and data center networks

Secure Access

- Dynamic Segmentation
- Campus- branch fabric EVPN/VXLAN
- NetConductor

Simplified Operations

- Single OS from access to core to DC
- Smart automation for flawless configs with NetEdit
- Built-in analytics with deep visibility and clear alerts
- Management flexibility (Cloud, on-prem, Web GUI, CLI, NetEdit, AFC)
- Unified cloud-based single pane of glass management for wired, WLAN, SD-WAN

HPE Aruba Networking CX Switches



Technology Innovation

- AOS-CX: Programmable, modern, designed to scale access to DC
- NAE: built-in analytics monitor, baseline, troubleshoot
- NetEdit: error-free automated rollouts
- VSX live upgrades, VSF rolling stack, & Hot Patching: HA during upgrades
- In-house ASICs: high-performance, programmable agility
- Central: Unified, AIOps, NetConductor

Faster Troubleshooting

- NAE built-in for monitoring, alerts, troubleshooting and root cause analysis
- Central integration with AI insights

Scalable, High-Performance Enterprise Switching

High-density fixed and modular switches with non-blocking architecture, speeds from 1G to 400G, extended table sizes, high power PoE, multi-gig, 80 Plus power supplies, fiber and copper transceivers and DACs

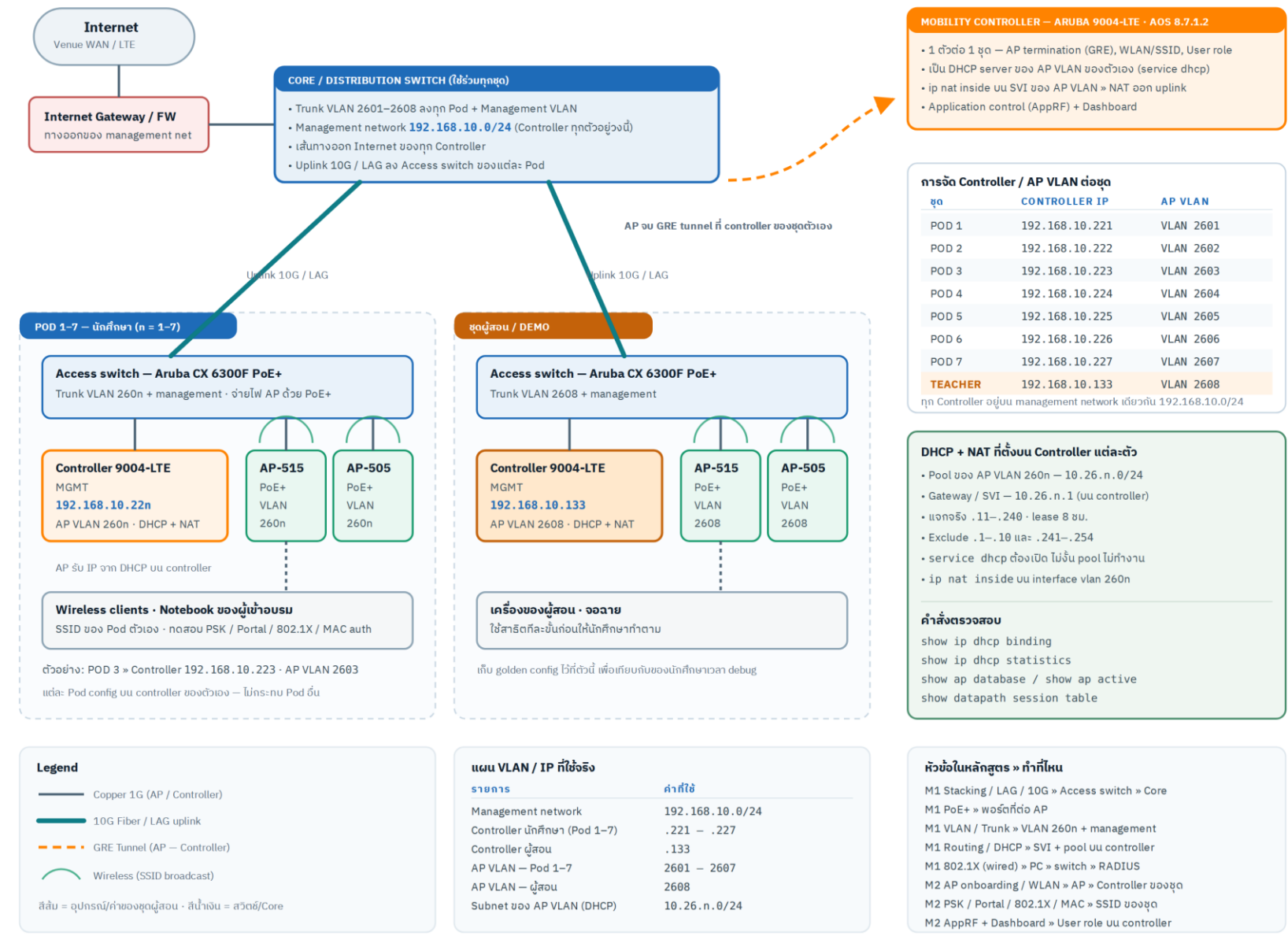
Customer First, Customer Last

- Industry leading warranty
- Global support and services
- Consumption choice: GL4A, NaaS
- Financing: HPEFS

Switch Labs

Aruba Campus Network Training – Lab Topology

RMUTT x Commserv Siam · 26 ส.ค. 2569 · Mobility Controller: Aruba 9004-LTE (ArubaOS 8.7.1.2) · 7 ชุดนักศึกษา + 1 ชุดผู้สอน



LAB 0 ตรวจสอบและหาชุดของตัวเอง

ที่โต๊ะ รู้ว่าชุดตัวเองคือหมายเลขอะไร ต่อ console ถูกเครื่อง และจดค่าที่ต้องใช้ทั้งวันไว้

1. ดูหมายเลขชุดที่โต๊ะ แล้วเปิดหัวข้อ Config ด้านบน เลือกแท็บของชุดตัวเอง — ค่าทุกตัวในวันนี้อ้างอิงจากตารางนั้น
2. ต่อสาย console เข้า Access switch (CX 6300F) ตั้ง terminal 115200 8-N-1 ไม่มี flow control
3. ต่อสาย console เข้า Controller (9004-LTE) อีกเส้น หรือจะรอเข้าผ่านเว็บก็ได้
4. จดไว้ให้เห็นตลอดวัน: Pod n · AP VLAN 260n · Controller MGMT 192.168.10.22n · Subnet 10.26.n.0/24

ตรวจสอบผล

```
show version
show system
```

ระวัง

ต่อ console ผิดเครื่องของชุดอื่นคือปัญหาที่เสียเวลาที่สุดของวัน — เช็ก hostname ก่อนพิมพ์อะไรเสมอ

LAB 1 Access switch — คำเริ่มต้นและ management

Access switch (CX 6300F) เข้าถึงสวิตช์ได้ทั้ง console และ SSH และตั้งชื่อให้รู้ว่าเป็นของชุดไหน

1. ตั้งรหัสผ่าน admin ตอน login ครั้งแรก
2. ตั้ง hostname ให้เป็นของตัวเอง

```
configure terminal
hostname POD1-SW
```
3. ตั้ง IP ของ management interface — คนละวงกับ AP VLAN และอยู่บน VRF mgmt

```
interface mgmt
no shutdown
ip static <IP ที่ผู้สอนแจก>/24
default-gateway <gateway ที่ผู้สอนแจก>
```
4. เปิด SSH บน VRF mgmt แล้วบันทึก

```
ssh server vrf mgmt
write memory
```

ตรวจสอบ

```
show interface mgmt
show running-config | include hostname
```

ระวัง

IP management ของสวิตช์ผู้สอนแจกให้ต่างหาก ไม่ใช่ 192.168.10.22n — เลขนั้นเป็นของ Controller

LAB 2 VLAN, Trunk ขึ้น Core และพอร์ต AP

Access switch (CX 6300F) สร้าง AP VLAN ของชุดตัวเอง ส่งขึ้น Core ได้ และเปิดพอร์ตที่ AP เลียบอยู่

1. สร้าง VLAN ของชุดตัวเอง

```
configure terminal
vlan 2601
name AP-POD1
```

2. ตั้งพอร์ต uplink ขึ้น Core เป็น trunk ที่ยอมให้ VLAN ของตัวเองผ่าน (ฝั่ง Core ผู้สอน config ไว้แล้ว)

```
interface 1/1/49
no shutdown
vlan trunk native 1
vlan trunk allowed 2601
```

3. ตั้งพอร์ตที่ต่อ AP เป็น access VLAN ของชุดตัวเอง

```
interface 1/1/1-1/1/2
no shutdown
vlan access 2601
```

4. บันทึก

```
write memory
```

ตรวจสอบผล

```
show vlan
show interface brief
show lldp neighbor-info
```

ระวัง

ลืม no shutdown คือสาเหตุอันดับหนึ่งที่ AP ไม่ขึ้น · ใส่ VLAN ผิดเลข AP จะไม่ได้ IP เพราะ DHCP อยู่บน controller ของ VLAN นั้น

LAB 3 PoE สำหรับ AP

Access switch (CX 6300F) ยืนยันว่าพอร์ตที่ต่อ AP จ่ายไฟจริง และอ่าน power budget เป็น

1. ดูสถานะ PoE ทั้งเครื่องและบัพไฟที่เหลือ
`show power-over-ethernet brief`
2. ดูรายพอร์ตว่าจ่ายไฟกี่วัตต์ คลาสอะไร
`show power-over-ethernet interface 1/1/1`
3. ถ้าต้องปิด-เปิดไฟพอร์ตเพื่อ reboot AP
`interface 1/1/1`
`no power-over-ethernet`
`power-over-ethernet`

ตรวจสอบ

```
show power-over-ethernet brief
```

ระวัง

AP-515 กิน PoE+ (คลาส 4) ถ้าเสียบ AP หลายตัวบนสวิตช์ที่
งบไฟไม่พอ ตัวหลังจะไม่บูต — อ่าน available power ก่อน
โทษ AP

LAB 4 LAG ขึ้น Core

Access switch (CX 6300F) รวมสองพอร์ต uplink เป็นลิงก์เดียวแบบ LACP (ฝั่ง Core ผู้สอนตั้งไว้แล้ว)

1. สร้าง LAG แล้วตั้งเป็น trunk เหมือน uplink เดิม

```
configure terminal
interface lag 1
  no shutdown
  vlan trunk native 1
  vlan trunk allowed 2601
  lacp mode active
```

2. ย้ายพอร์ตจริงสองพอร์ตเข้า LAG

```
interface 1/1/49-1/1/50
  no shutdown
  lag 1
```

3. บันทึก

```
write memory
```

ตรวจสอบผล

```
show lacp interfaces
show lacp aggregates
show interface lag 1
```

ระวัง

ถ้าฝั่ง Core เป็น static แต่ฝั่งเราเป็น LACP ลิงก์จะไม่ขึ้น —
ถามผู้สอนก่อนว่าฝั่งโน้นตั้งไว้แบบไหน

ArubaOS-CX switch CLI contexts

Operator (>)

- Enables execution of commands to view—but not change—the configuration
- Switch prompt:
 - switch>

Manager (#)

- From manager context (#), execute commands that do not require saving changes to configuration
- Switch prompt:
 - switch#

Global configuration (config)#

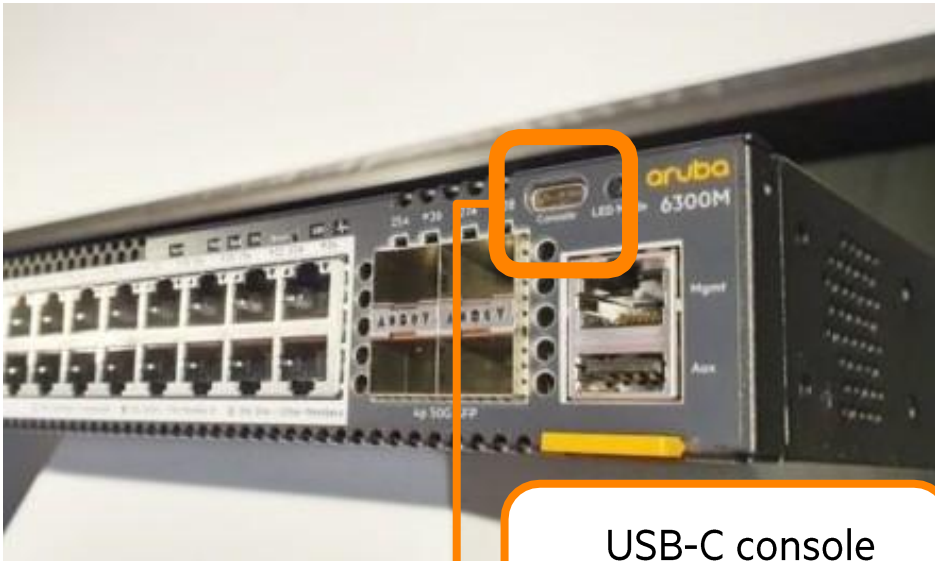
- From global configuration context (config), execute commands that change configuration of switch
- Switch prompt:
 - switch(config)#

Other configuration

- switch(config-if)#
- switch(config-router)#
- switch(config-vlan-100)#

Note: If you log into a device with the admin role you are immediately put into enable context. Type 'exit' at enable to disconnect (not be put into operator mode).

Connecting to Console Port



USB-C console
No RJ45 console
(CX6300)



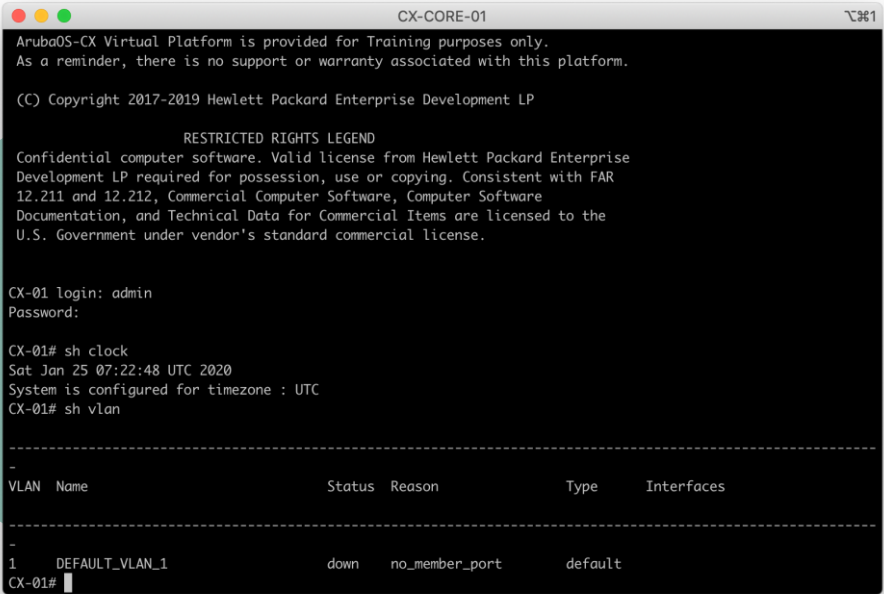
CX Switch

Direct serial connection to console port

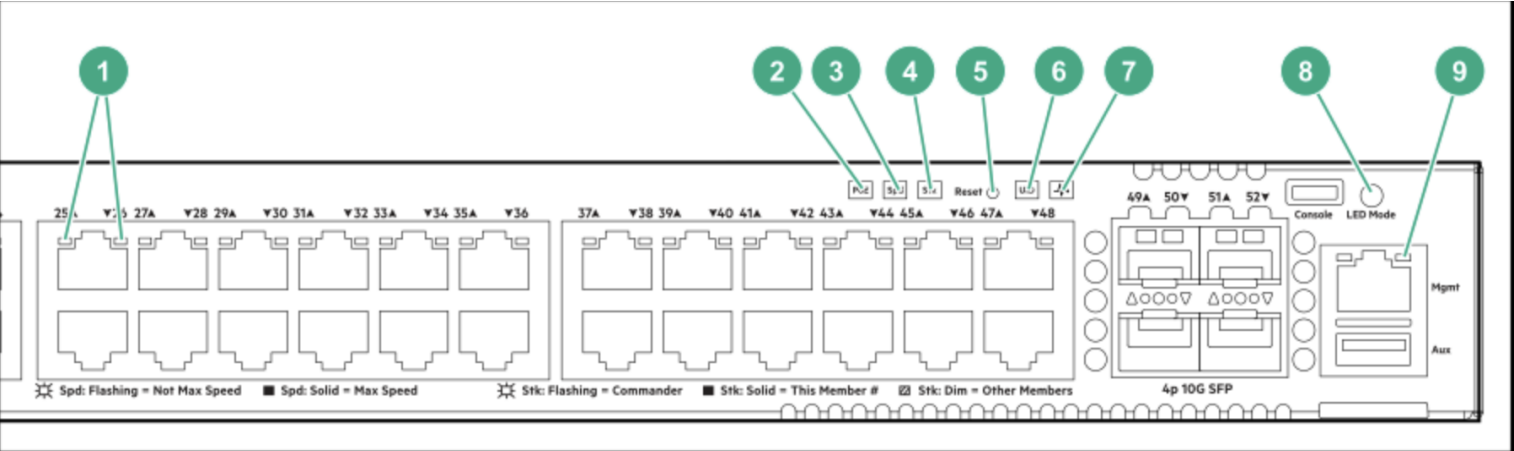
CX Switch	Serial Console Type
8400, 8325-48Y8C	RJ45 and Micro USB-B
8320, 8325-32C	RJ45
6400	RJ45 and USB-C
6300	USB-C
6200 (12 port)	USB-C, RJ45

Terminal software settings

- Speed: 115200 bps
- Data bits: 8
- Stop bits: 1
- Parity: None
- Flow control: None



Switch LEDs



Label	Description
1	Switch port LEDs
2	PoE LED (only support POE)
3	Speed mode selected LED
4	Stk LED
5	Reset button
6	UID (Unit Identification)
7	Global Status LED
8	Mode select button
9	OOBM port LED

User Accounts

- admin is the default user with **no password**
- Administrators
 - Users with administrator rights can execute any command.
 - Default setting will have admin user account without password, and you are required to enter a **new password after login**
- Operators
 - Users with operator rights can execute commands in the operator context (>) only
- Auditors
 - Users with auditor rights can execute commands in the auditor context (**auditor>**) only
 - Limited view of show command

Lab 1.1 – Initialize Admin Account and Change Hostname

```
switch login: admin  
Password:
```

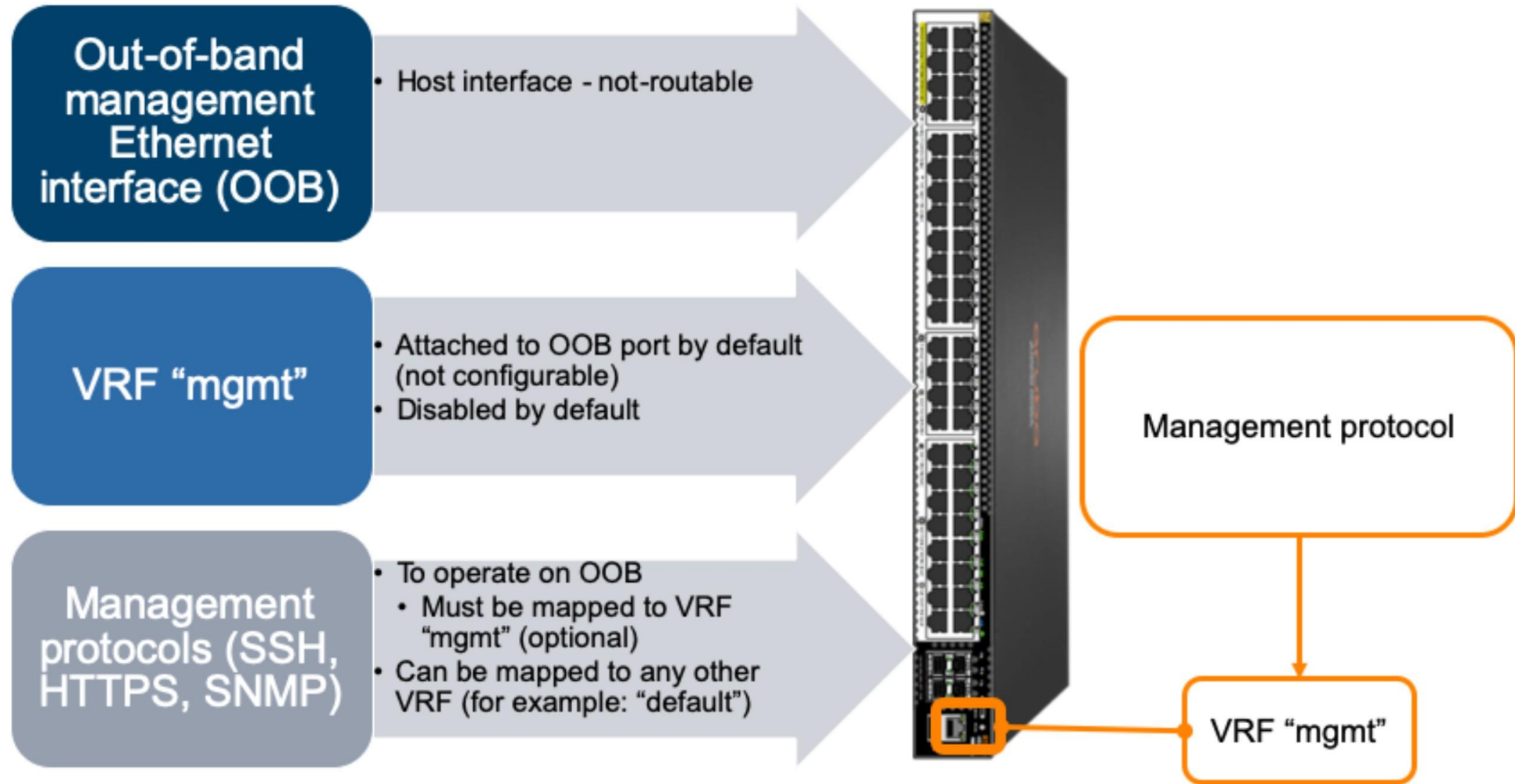
```
Please configure the 'admin' user account  
password. Enter new Password: ****  
Confirm new password: ****
```

1. Default setting will have admin user account without password, and you are required to enter a new password after login
2. Enter any password string you want, we will change to the new one in the next lab

```
Switch# config terminal  
Switch(config)# hostname CX-CORE-01  
CX-CORE-01(config)#
```

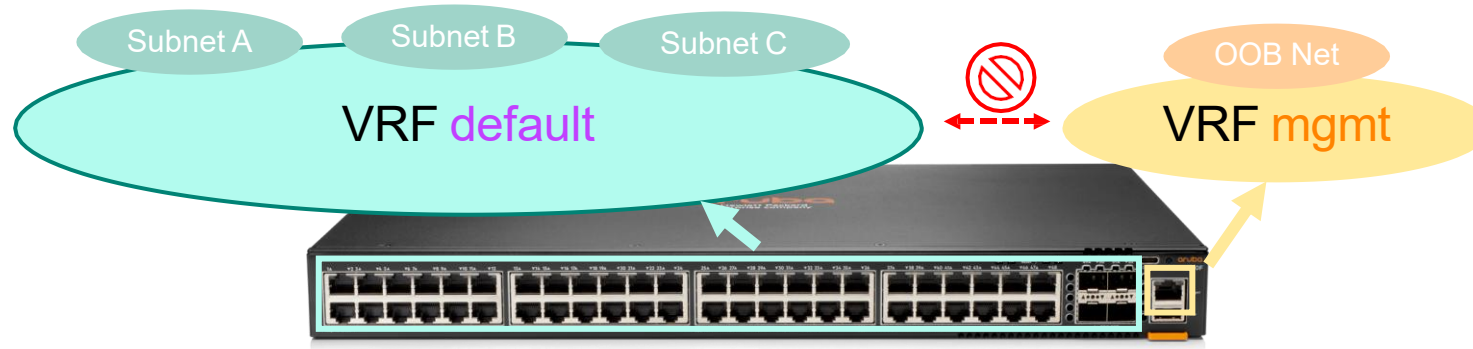
1. Change switch name to CX-CORE-01

Out of band Management



VRF-lite in ArubaOS-CX

- VRF technology is an integral part of the operating system
- There is two default VRF called default and mgmt
 - The management interface belongs to the **mgmt**
 - All data L3 interfaces belong to the **default**



HTTPS Server - WebUI

Requires the admin user to have a password

Must be attached to a VRF

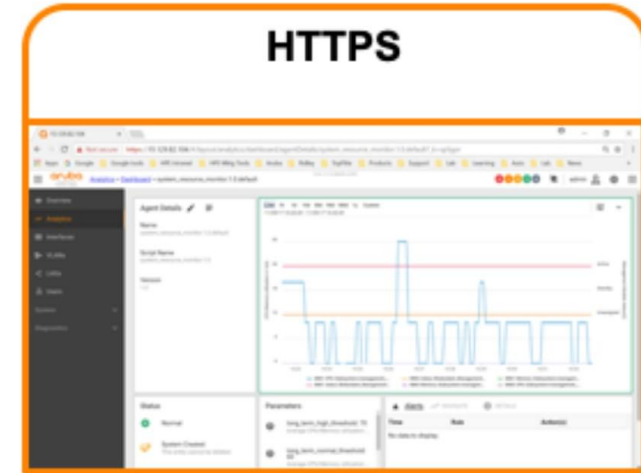
- “mgmt” for OOB management
- “default”
- Any other

Can be assigned simultaneously to multiple VRFs

API is accessible in read-only or read-write

Web-UI is disabled by default

Note: HTTP server is not supported



VRF “mgmt”

REST Interface

Requires admin user to have a password

REST reached via HTTPS

Access can be read-write or read-only (default read-only)



REST

ArubaOS-CX REST API

RESTful interface for ArubaOS-CX switch software

AAA_Server_Group	Show/Hide	Lock Operations	Expand Operations
AAA_Server_Group_Profile	Show/Hide	Lock Operations	Expand Operations
ACL	Show/Hide	Lock Operations	Expand Operations
ACL_Entry	Show/Hide	Lock Operations	Expand Operations
Aggregate_address	Show/Hide	Lock Operations	Expand Operations
SGP_Path_Filter	Show/Hide	Lock Operations	Expand Operations
SGP_Path_Filter_Entry	Show/Hide	Lock Operations	Expand Operations
SGP_Community_Filter	Show/Hide	Lock Operations	Expand Operations
SGP_Community_Filter_Entry	Show/Hide	Lock Operations	Expand Operations
SGP_Neighbor	Show/Hide	Lock Operations	Expand Operations
SGP_Route	Show/Hide	Lock Operations	Expand Operations
SGP_Router	Show/Hide	Lock Operations	Expand Operations

VRF "mgmt"

Lab 1.3 - Enable Remote Management Access

```
CX-CORE-01# configure terminal
CX-CORE-01(config)# https-server vrf mgmt
CX-CORE-01(config)# ssh server vrf mgmt
CX-CORE-01(config)# https-server rest access-mode read-only
CX-CORE-01(config)# interface mgmt
CX-CORE-01(config-if-mgmt)# ip static 10.1.10.101-103/24
CX-CORE-01(config-if-mgmt)# default-gateway 10.1.10.1
CX-CORE-01(config-if-mgmt)# nameserver 8.8.8.8
CX-CORE-01(config-if-mgmt)# no shutdown
```

- 1. Bind the https and ssh server with VRF “mgmt.”
- 2. Assign IP address, Gateway and DNS to the management interface

- 1. Open the browser and see if you can access CX switches
- 2. SSH CX switches and make sure you have the access
- 3. Ping to the switch IP address on mgmt VRF

MGMT IP Address
Group1 - CX-CORE = 10.1.10.101-103/24
Group2 - CX-CORE = 10.1.10.104-106/24
Group3 - CX-CORE = 10.1.10.107-109/24
Group4 - CX-CORE = 10.1.10.110-112/24
Group5 - CX-CORE = 10.1.10.113-115/24
Group6 - CX-CORE = 10.1.10.116-118/24
Group7 - CX-CORE = 10.1.10.119-121/24

ping 10.1.10.101
PING 10.1.10.101 (10.1.10.101) 100(128) bytes of data.

108	bytes	from	10.1.10.101:	icmp_seq=1	ttl=128	time=1.04 ms
108	bytes	from	10.1.10.101:	icmp_seq=2	ttl=128	time=0.454 ms
108	bytes	from	10.1.10.101:	icmp_seq=3	ttl=128	time=0.763 ms
108	bytes	from	10.1.10.101:	icmp_seq=4	ttl=128	time=0.681 ms
108	bytes	from	10.1.10.101:	icmp_seq=5	ttl=128	time=0.604 ms

Lab 1.4 – Configure Clock, Timezone, DNS and SNMP

- Some of the basic configuration like DNS and Logging supports VRF binding

```
CX-CORE-01 # configure terminal
CX-CORE-01(config)# clock timezone Asia/Bangkok
CX-CORE-01(config)# ip dns server-address 8.8.8.8 vrf mgmt
CX-CORE-01(config)# ntp server 0.th.pool.ntp.org
CX-CORE-01(config)# ntp vrf mgmt
CX-CORE-01(config)# ntp enable
CX-CORE-01(config)# snmp-server system-description CX-CORE-01
CX-CORE-01(config)# snmp-server system-contact Administrator
CX-CORE-01(config)# snmp-server system-location CX_LAB
CX-CORE-01(config)# snmp-server community cx_snmp
CX-CORE-01(config)# snmp-server vrf mgmt
CX-CORE-01(config)# snmp-server host x.x.x.x trap version v2c community cx_snmp
CX-CORE-01(config)# logging x.x.x.x vrf mgmt
```

- SNMP trap and log server are dummy IP address. It is intended for student to practice how to config these two settings.

```
CX-CORE-01# sh ntp associations
```

ID	NAME	REMOTE	REF-ID	ST	LAST	POLL	REACH
* 1	0.th.pool.ntp.org	202.12.97.45	.PPS.	1	7	64	377

```
CX-CORE-01# show clock
Tue Jan 28 13:50:39 ICT 2020
System is configured for timezone : Asia/Bangkok
```

- Verify date, time and timezone

Configuration Management

- “**write memory**” to save running configuration into the Startup configuration
- **Types of checkpoints**
 - **System generated checkpoints:** Checkpoints that the system automatically generates whenever a configuration change is detected.
 - **User generated checkpoints:** Checkpoints that the user manually generates
- Total limit of checkpoint is **64** including the startup configuration; **32** each type
- After the limit the system will overwrite the oldest checkpoint
- For each additional configuration change, the **300** second timeout counter is reset. Once the timeout limit is reached with no additional configuration changes detected, the system will automatically generate a new checkpoint
- This system generated checkpoints are listed with a time stamp format CPC<**YYYYMMDDHHMMSS**>
- The checkpoints can be applied using the “**checkpoint rollback**” or “**copy checkpoint**” command
- **Difference** between two configuration can be viewed between two different checkpoints or between any checkpoint and the startup/running configuration

Lab 1.5 – Creating Configuration Checkpoint

```
CX-CORE-01 # copy running-config checkpoint initial_config  
Large configuration changes will take time to process, please be patient.
```

```
CX-CORE-01# show checkpoint  
CPC20200128072408  
CPC20200128075706  
initial_config
```

- Use this command to show all saved configuration checkpoint

```
CX-CORE-01# show checkpoint initial_config  
.  
.  
.
```

- Use this command to show initial_config checkpoint

Lab 1.7 – Checkpoint Rollback

Use the **same CX switch in lab 1.6** to do this lab

```
CX-CORE-01# sh session-timeout
session-timeout: 0 minute
```

```
CX-CORE-01# sh banner motd
#####
TEST BANNER
#####
```

```
CX-CORE-01# sh snmp community
-----
SNMP communities
-----
public
```

1. Record the show output result and verify the difference after perform a configuration rollback

2. After that, issue a rollback command. You have two way to rollback your configuration to any checkpoint
3. Verify the show out result after rollback your configuration. What is the difference?

```
CX-CORE-01# copy checkpoint initial_config running-config
Configuration changes will take time to process, please be patient.
```

Option 1

```
CX-CORE-01# checkpoint rollback initial_config
Configuration changes will take time to process, please be patient.
```

Option 2

Knowledge Check #1

Which console interfaces and default baud rates are supported on the HPE Aruba CX 6200F 12-port model? (Select two.)

- a. USB-A, baud rate = 115200
- b. USB-C, baud rate = 9600
- c. RJ-45, baud rate = 115200
- d. USB-C, baud rate = 115200
- e. RJ-45, baud rate = 112500

Knowledge Check #2

Does the HPE Aruba 6200F model support only two VRF?

- a. True.
- b. False.

Knowledge Check #3

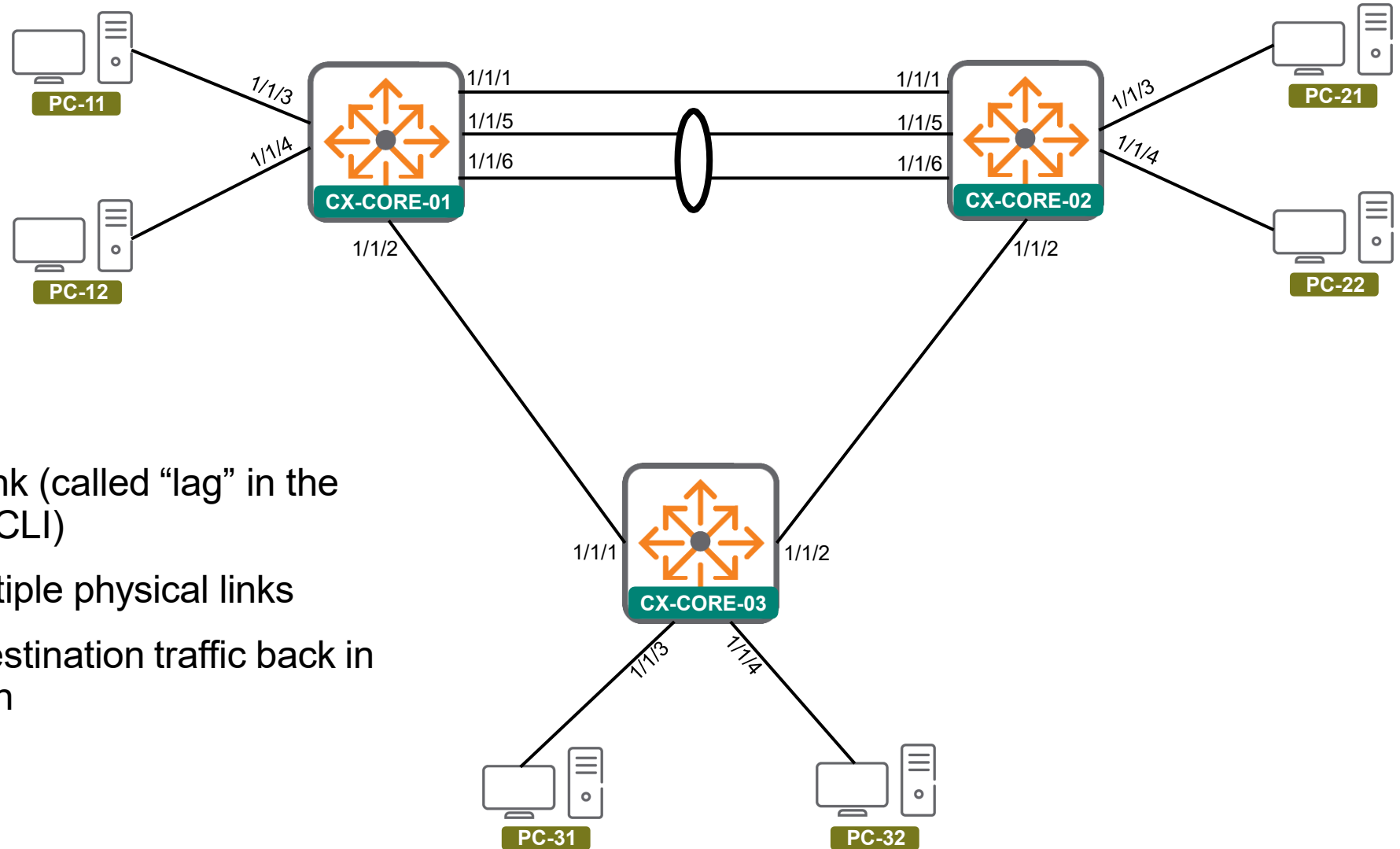
The REST API on Aruba CX supports both read and write operations (GET, POST, PUT, DELETE)?

- a. True.
- b. False.

Lab 2

Layer 2 Switching and Features

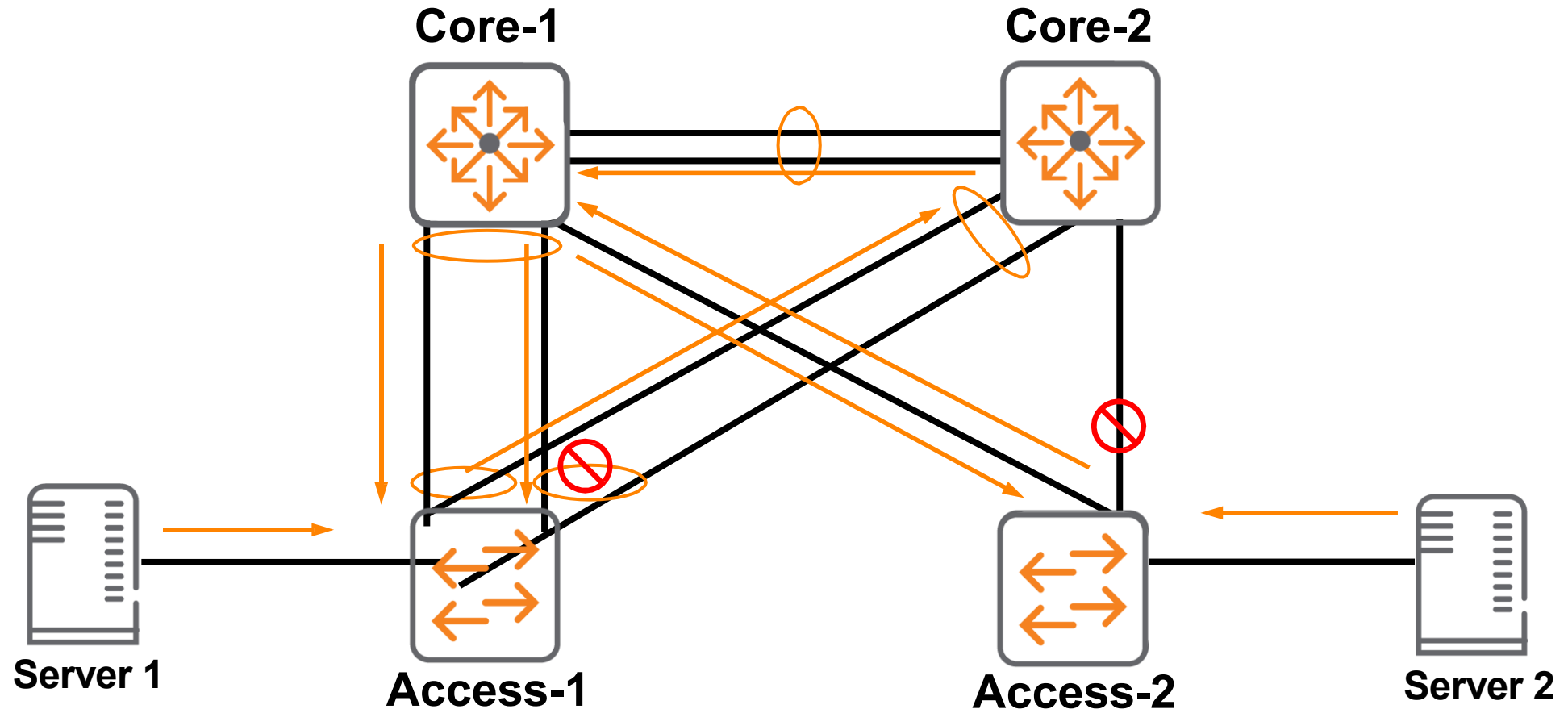
Link Aggregation



- Acts as one logical link (called “lag” in the ArubaOS-CX switch CLI)
- Carries traffic on multiple physical links
- Never floods multi-destination traffic back in the same aggregation

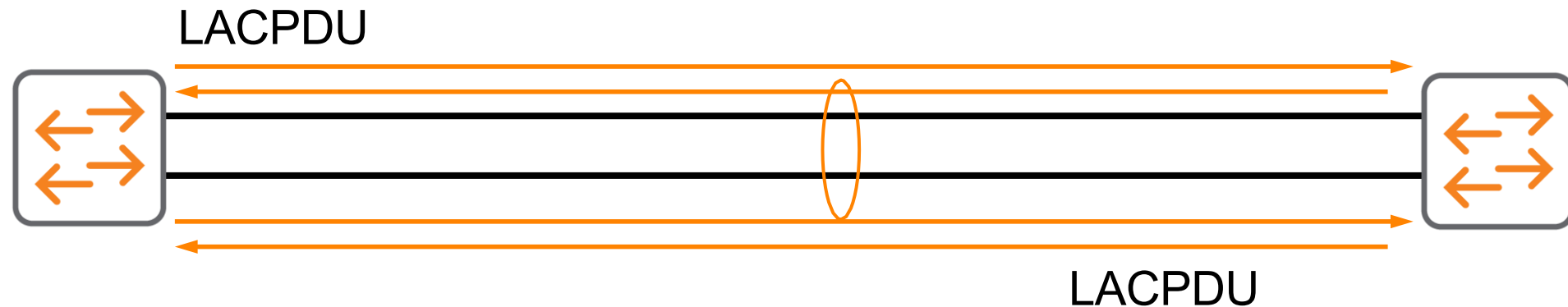
Potential issue with manual link aggregation

- What if you accidentally configure the wrong ports in a link aggregation?



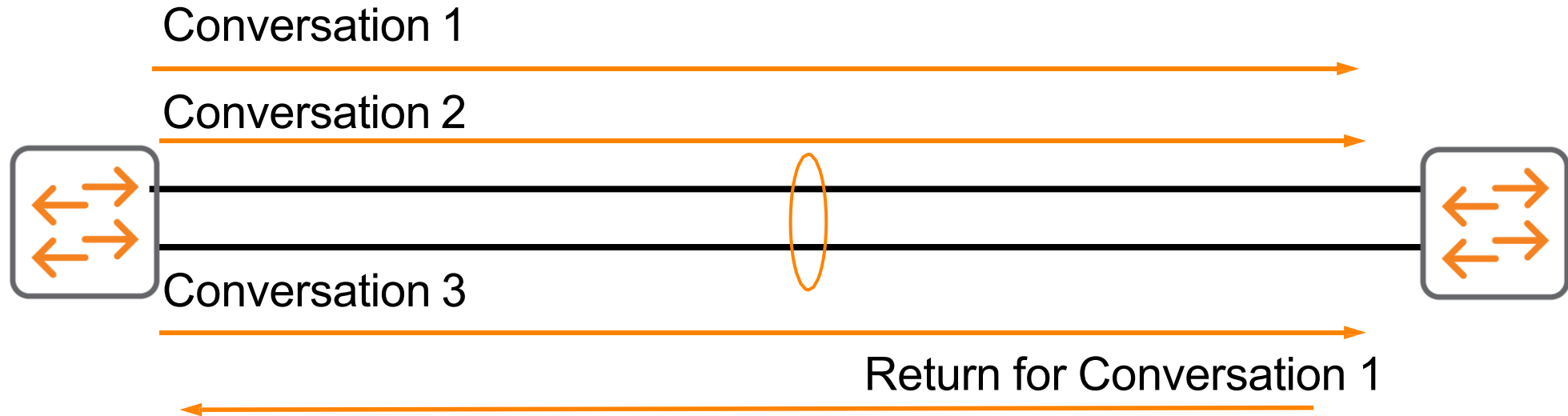
LACP

- Uses messages to establish the link aggregation
- Ensures that all links are connected to the same peer (system ID) and link aggregation (operational key)
- Ensures compatibility for other settings
- Manages adding and removing links

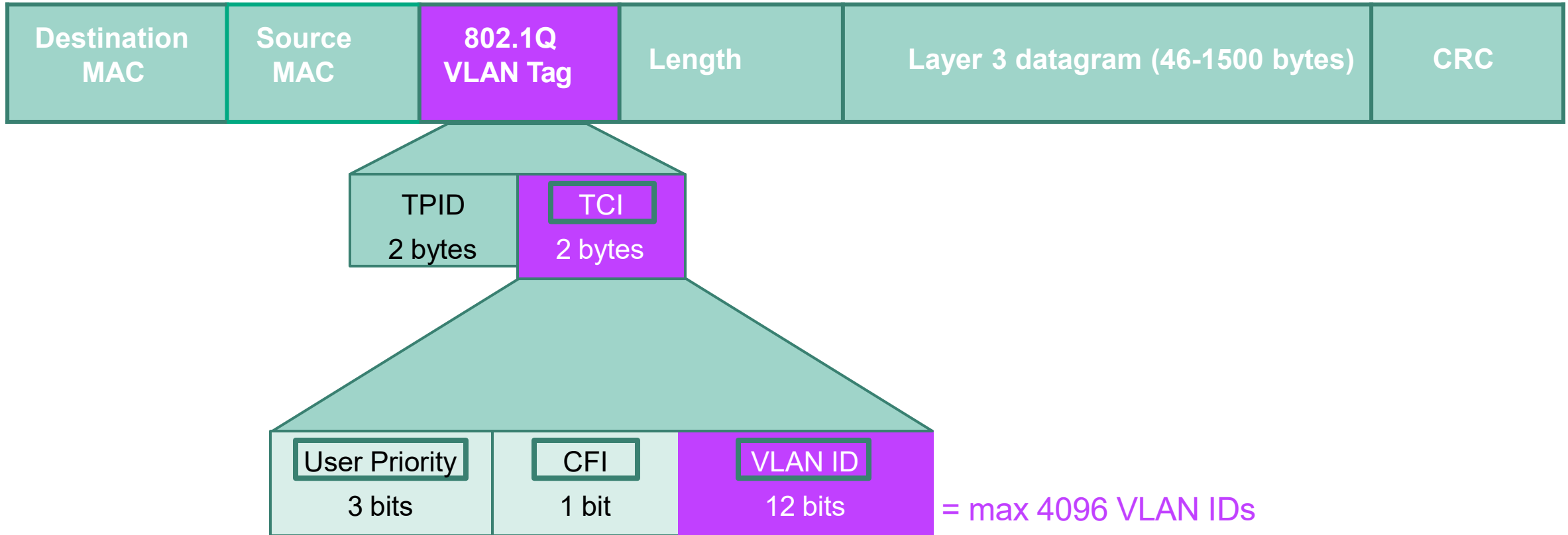


Load-sharing traffic over a link aggregation

- The link aggregation uses a hash to select a link for each *conversation*:
 - Conversation defined by the load-sharing mode
 - MAC source and destination address (L2-based)
 - IP source and destination address (L3-based) = Default option
 - Source and destination UDP/TCP ports (L4-based) = Only 8400 series switches support L4 hashing
 - Link for a conversation stays constant
 - Traffic tends to be balanced more evenly as conversations increase



Ethernet Frame Structure



Knowledge Check

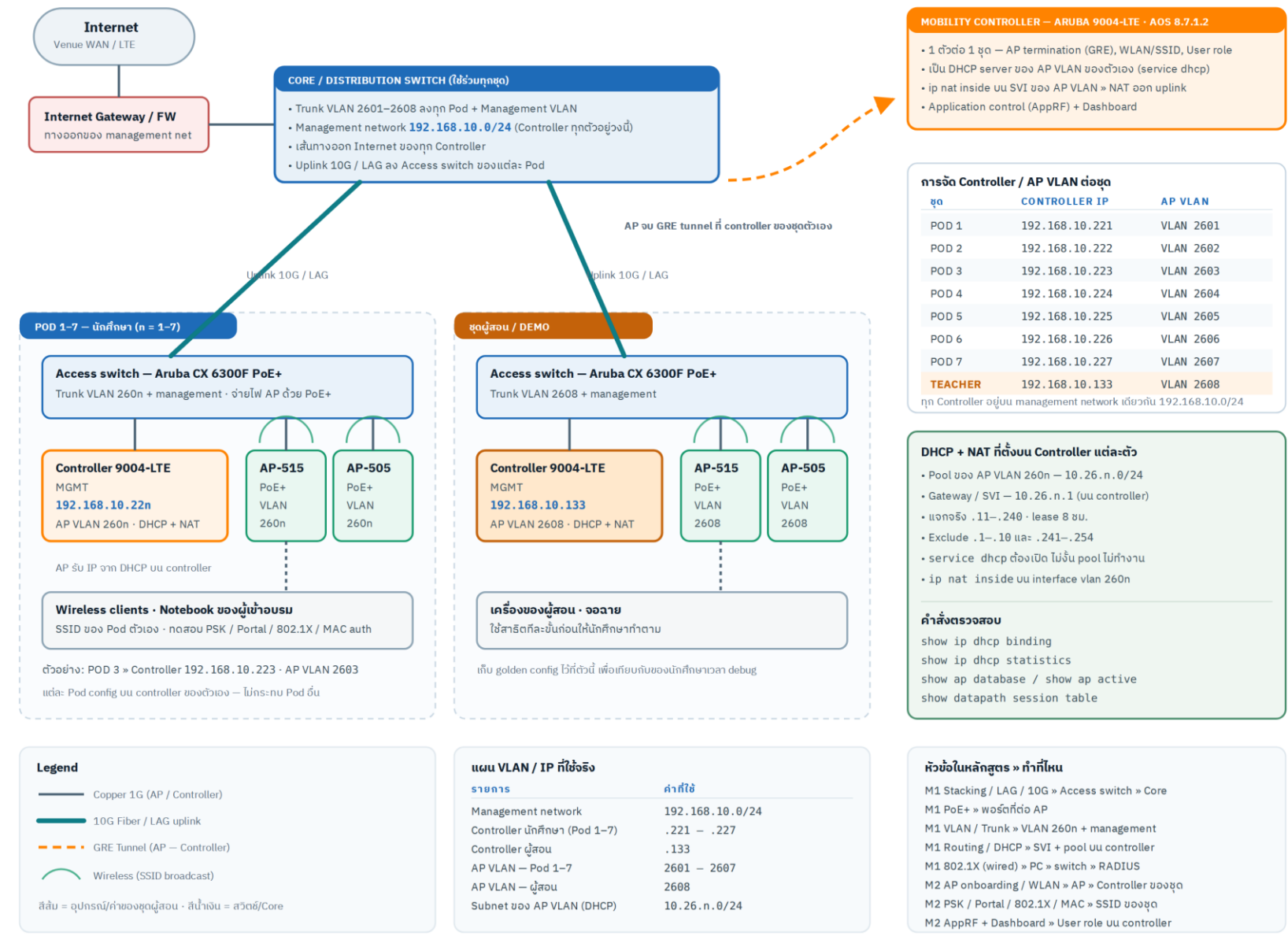
What feature allows combining multiple physical links into a single logical interface?

- a. VLAN
- b. STP
- c. LACP (Link Aggregation Control Protocol)
- d. LLDP

WLAN Labs

Aruba Campus Network Training – Lab Topology

RMUTT x Commserv Siam · 26 ส.ค. 2569 · Mobility Controller: Aruba 9004-LTE (ArubaOS 8.7.1.2) · 7 ชุดนักศึกษา + 1 ชุดผู้สอน



LAB 5 Controller — ค่าเริ่มต้น, license, เวลา

Controller (9004-LTE) ทำให้ controller ของชุดตัวเองขึ้นมาและเข้าเว็บได้

1. บูดครั้งแรกแล้วตอบ full-setup ที่ console — ตั้ง system name, country code TH, IP ของ controller
2. เข้าเว็บ <https://192.168.10.221:4343> (เลขตามชุดตัวเอง) ผู้ใช้ admin
3. ใส่ license AP และ PEF ที่ Configuration → License
4. ตั้งเวลาและ timezone Asia/Bangkok ที่ Configuration → System → General
5. เปิด CPSec แบบ auto cert provisioning ที่ Configuration → System → Certificates

ตรวจสอบผล

```
show switchinfo  
show license  
show clock
```

ระวัง

เวลาผิดพลาดให้ certificate ของ AP ไม่ผ่าน แล้ว AP จะไม่ยอมขึ้น — ตั้งเวลาก่อนเอา AP เข้าเสมอ

LAB 6 Controller — VLAN, SVI, DHCP และ NAT

Controller (9004-LTE) ทำให้ controller เป็น gateway และ DHCP server ของ AP VLAN ของตัวเอง แล้ว NAT ออกเน็ต

1. เปิดหัวข้อ Config ด้านบน เลือกแท็บชุดตัวเอง กดคัดลอก แล้ววางลง CLI ของ controller
2. อ่านทีละท่อนว่าแต่ละบล็อกทำอะไร — VLAN กับ SVI ที่มี ip nat inside → ช่วง IP ที่กันไม่ให้แจก → DHCP pool → service dhcp
3. บันทึก
write memory

ตรวจสอบผล

```
show ip interface brief  
show ip dhcp binding  
show ip dhcp statistics
```

ระวัง

service dhcp ไม่เปิด pool จะไม่ทำงานแม้ config ถูกหมด ·
เพลอวางของชุดอื่นให้ลบด้วย no interface vlan และ no ip
dhcp pool ก่อนวางใหม่

LAB 7 AP onboarding

Controller (9004-LTE) + AP เอา AP สองตัวขึ้นมากอยู่กับ controller ของชุดตัวเอง และแยกเป็น AP group ของตัวเอง

1. เสียบ AP เข้าพอร์ตที่ทำไว้ใน LAB 2 — AP รับ IP จาก DHCP บน controller
2. ถ้า AP เคยใช้ที่อื่นมาก่อน ให้ reset โดยกดปุ่ม reset ค้างเกิน 10 วินาทีตอนเปิดเครื่อง
3. ชี้ AP มาที่ controller (convert to campus mode) ด้วย IP ของชุดตัวเอง
4. อนุมัติ AP ที่ Configuration → Access Points → Allowlist → Approve
5. สร้าง AP group ของชุดตัวเอง แล้วย้าย AP ทั้งสองตัวเข้ากลุ่ม

ตรวจสอบผล

```
show ap database
show ap active
show ap allowlist
```

ระวัง

AP ไม่ขึ้นให้ไล่จากล่างขึ้นบน — พอร์ตมีไฟ PoE ใหม่ → ได้ IP ใหม่ (show ip dhcp binding) → เห็นใน show ap database ใหม่

LAB 8 WLAN — PSK และ MAC Authentication

Controller (9004-LTE) สร้าง SSID แรกด้วย PSK แล้วเพิ่มการตรวจ MAC จากฐานข้อมูลในตัว controller

1. สร้าง WLAN ที่ Configuration → WLANs → + ตั้งชื่อ SSID เป็นของชุดตัวเอง เช่น POD1-IOT
2. เลือก AP group ของตัวเอง · VLAN 2601 · Forwarding mode Tunnel
3. ตั้ง security เป็น Personal (PSK) แล้วทดสอบด้วยโน้ตบุ๊ก
4. เพิ่ม MAC ของเครื่องทดสอบเป็น local user ที่ Configuration → Authentication → Internal Server → Users โดยใช้ MAC เป็นทั้ง username และ password
5. เปิด MAC authentication บน WLAN เดิม แล้วทดสอบใหม่

ตรวจสอบผล

```
show user-table  
show ap association
```

ระวัง

ใส่ MAC ผิดฟอร์แมตคือปัญหาที่เจอบ่อย — ดู MAC จริงจาก show user-table ของเครื่องที่เพิ่งต่อ แล้วคัดลอกไปใส่

LAB 9 WLAN — Captive portal สำหรับ Guest

Controller (9004-LTE) สร้าง SSID สำหรับผู้มาเยือน ที่ต้องล็อกอินผ่านหน้าเว็บก่อนใช้งาน

1. สร้าง WLAN ใหม่ ชื่อ POD1-GUEST · VLAN 2601 · AP group ของตัวเอง
2. เลือก security เป็น Captive portal (Guest)
3. สร้าง guest user ใน Internal Server
4. ต่อด้วยมือถือ — ต้องตั้งหน้า login ก่อนถึงจะออกเน็ตได้

ตรวจสอบผล

```
show user-table  
show rights
```

ระวัง

ไม่ตั้งหน้า login ให้ดูว่าเครื่องได้ IP กับ DNS จริงหรือยัง —
captive portal ต้องให้ DNS ผ่านก่อน

LAB 10 WLAN — 802.1X ด้วย Internal Server

Controller (9004-LTE) สร้าง SSID ที่ผู้ใช้ต้องใส่ username และ password ของตัวเอง โดยตรงกับฐานข้อมูลในตัว controller

1. สร้าง WLAN ใหม่ ชื่อ POD1-EMP · VLAN 2601 · AP group ของตัวเอง
2. เลือก security เป็น Enterprise (802.1X) แล้วเลือก auth server เป็น Internal
3. สร้าง user ที่ Configuration → Authentication → Internal Server → Users
4. ต่อด้วยโน้ตบุ๊ก ใส่ username และ password ที่สร้างไว้

ตรวจสอบผล

```
show user-table  
show aaa authentication-server internal
```

ระวัง

งานนี้ไม่มี ClearPass — 802.1X จึงทำได้เฉพาะฝั่ง WLAN ที่ controller ตรวจสอบเอง ฝั่ง wired บนสวิตช์ทำไม่ได้เพราะไม่มี RADIUS ให้ใช้

LAB 11 Role, Policy และ Dashboard

Controller (9004-LTE) จำกัดสิทธิ์ผู้ใช้ด้วย role แล้วดูผลบน dashboard

1. สร้าง network alias ของวงที่จะห้าม ที่ Configuration → Roles & Policies → Aliases
2. สร้าง policy ที่ deny ปลายทางเป็น alias นั้น แล้ว permit ที่เหลือ
3. สร้าง role ใหม่ แล้วผูก policy เข้าไป
4. ผูก role ให้ user ที่สร้างไว้ใน LAB 10 แล้วต่อใหม่
5. ดูผลที่ Dashboard → Clients และดู AppRF ว่าเห็น traffic อะไรบ้าง

ตรวจสอบผล

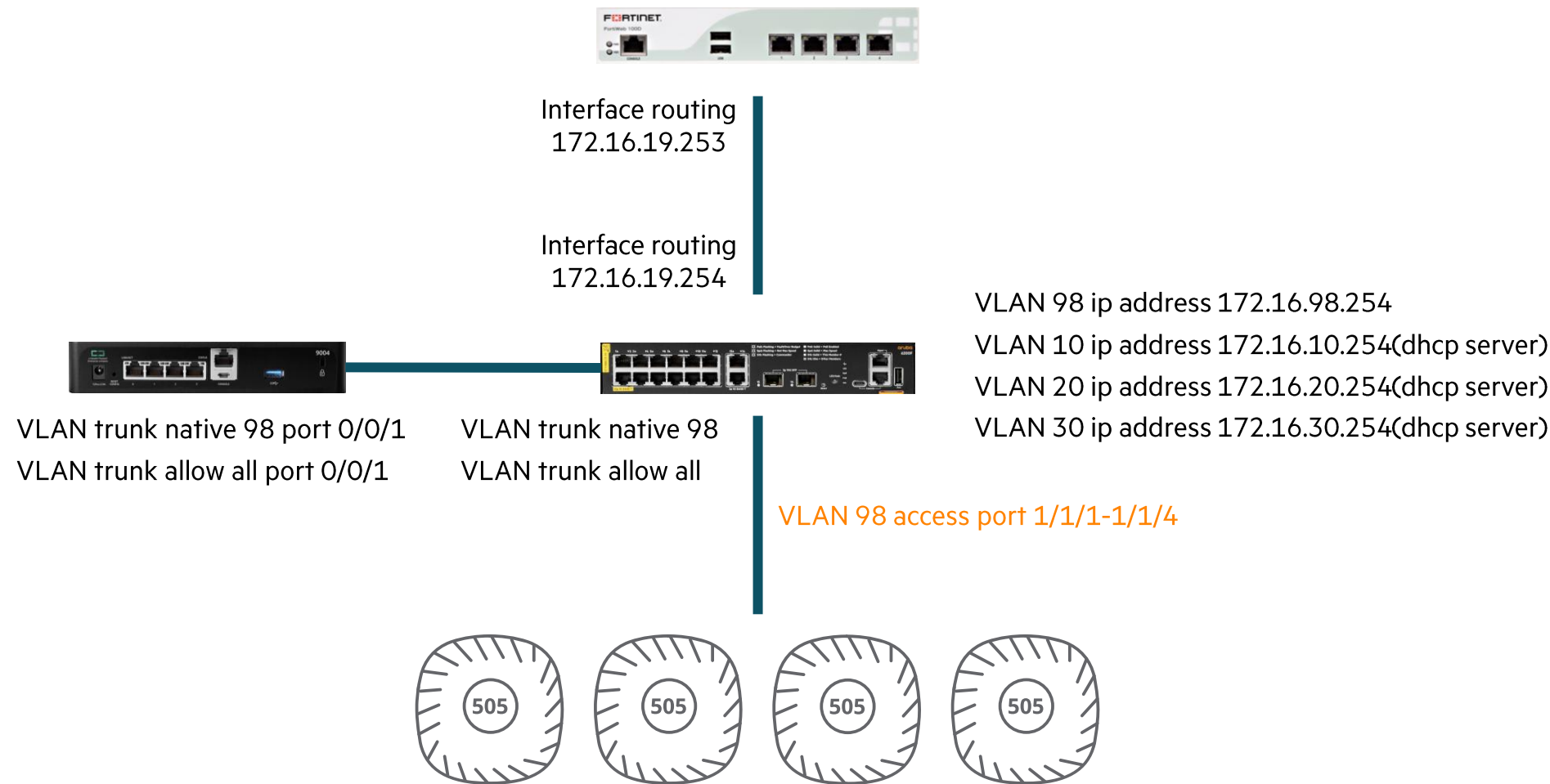
```
show user-table  
show rights  
show datapath session table
```

ระวัง

role ที่ผูกกับ user มีผลตอนต่อใหม่เท่านั้น — ตัดการเชื่อมต่อแล้วต่อใหม่ก่อนสรุปว่า policy ไม่ทำงาน

A.Diagram

Diagram



B.Console Cable

C.Initial Mobility Controller(MC)

- **Initial Controller**
- **Generate trail license AP & PEF**
- **Add License AP & PEF**
- **Set Time**
- **Configure CPSec**

Enter initial command : Full-setup

```
Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning...
'enable-debug'      : Enable auto-provisioning debug logs
'disable-debug'     : Disable auto-provisioning debug logs
'mini-setup'        : Start mini setup dialog. Provides minimal customization and requires DHCP server
'full-setup'         : Start full setup dialog. Provides full customization
'static-activate'    : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): full-setup

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): yes
```

Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning....

- 'enable-debug' : Enable auto-provisioning debug logs
- 'disable-debug' : Disable auto-provisioning debug logs
- 'mini-setup' : Start mini setup dialog. Provides minimal customization and requires DHCP server
- ' **full - setup** ' : Start full setup dialog. Provides full customization
- 'static-activate' : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): **full-setup**

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): **yes**

Initial MC-0X

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

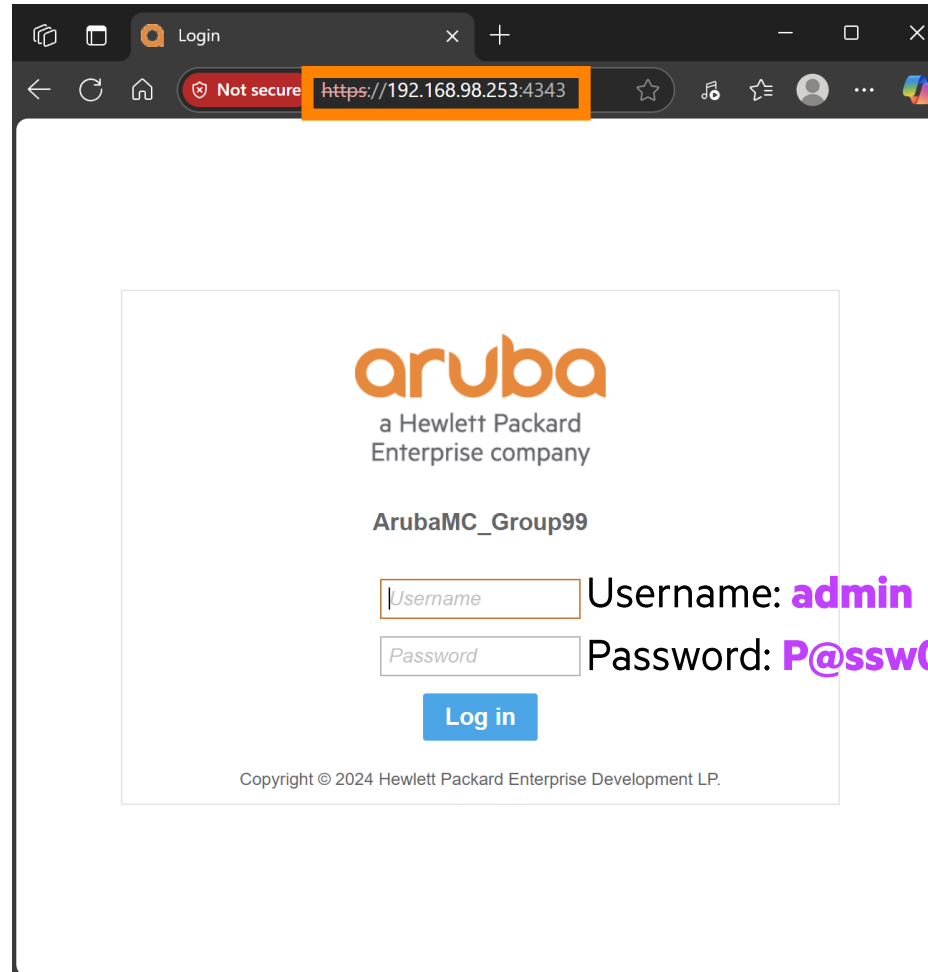
```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_Group99
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]:
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 192.168.98.253
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 192.168.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los_Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00
Enter Date (MM/DD/YYYY) [6/14/2025]:
Enter Password for admin login (up to 32 chars): *****
Re-type Password for admin login: *****
```

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_GroupXX
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]: [Enter]
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 172.16.98.1-7 //เรียงตาม group
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 172.16.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00 //Current Time HH:MM:SS
Enter Date (MM/DD/YYYY) [6/14/2025]: [Enter] // Check Date Current
Enter Password for admin login (up to 32 chars): P@ssw0rd
Re-type Password for admin login: P@ssw0rd
```

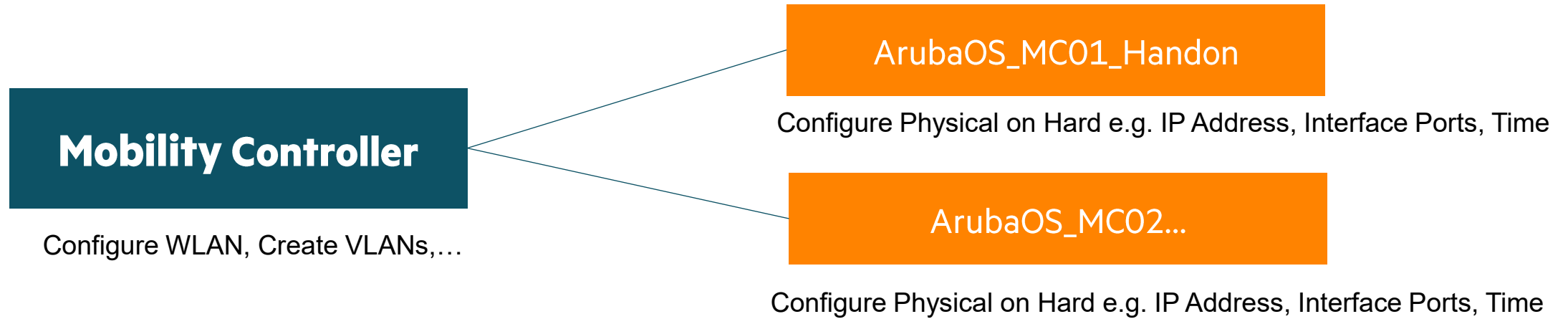
Add License

Enter Web Browser URL >> [https://172.16.98.\(1-7\):4343](https://172.16.98.(1-7):4343)



The screenshot shows a web browser window with the title "Login". The address bar displays "https://192.168.98.253:4343" with a "Not secure" warning. The main content area features the Aruba logo, the text "a Hewlett Packard Enterprise company", and the identifier "ArubaMC_Group99". Below this, there are input fields for "Username" and "Password". To the right of the "Username" field, the text "Username: admin" is displayed. To the right of the "Password" field, the text "Password: P@ssw0rd" is displayed. A blue "Log in" button is positioned below the password field. At the bottom of the page, a copyright notice reads "Copyright © 2024 Hewlett Packard Enterprise Development L.P."

Hierarchy



Add License

aruba MOBILITY CONTROLLER ArubaMC_Group99

ACCESS POINTS 0 0 CLIENTS 0 0 0 ALERTS 0

admin

1 Mobility Controller

Dashboard

2 Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

3 License

Redundancy

Connect to external license server: ☐

▼ Usage

Usage Summary	Usage by This Controller		Usage by Other Controller			
	AP Access Points	PEF Policy Enforcement Firewall	RF Protect Wireless Intrusion Protection	ACR Advanced Cryptography	WebCC Web Content Classification	VIA Virtual Intranet Access
Feature Enabled	☒	☐	☐	☐	☐	☒
Scope	Per-AP	Per-AP	Per-AP	Per-Session	Per-AP	Per-Session
Licenses Installed	0	0	0	0	0	0
Expired Licenses	0	0	0	0	0	0
Active Licenses	0	0	0	0	0	0
Licenses Used	0	0	0	0	0	0
Licenses Remaining Available	0	0	0	0	0	0

4 > Inventory

Set Time

1

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

2

ArubaMC_Group99

Admin AirWave CPSec Certificates SNMP Logging Profiles

Basic Info

Password for user admin:

Retype password:

> Domain Name System

CPSec with auto cert provisioning

The screenshot displays the Aruba Mobility Controller web interface. The interface is divided into a left sidebar and a main content area. The sidebar contains a menu with categories: **Mobility Controller** (1), **Configuration** (2), and **System** (3). Under **Configuration**, there are links for WLANs, Roles & Policies, Access Points, AP Groups, Authentication, Services, and Interfaces. Under **System**, there are links for Tasks, License, and Redundancy. The main content area has a top navigation bar with tabs: General, Admin, AirWave, **CPsec** (4), Certificates, SNMP, Logging, and Profiles. A **Pending Changes** button with a refresh icon (6) is located in the top right corner. The **CPsec** tab is active, showing the **Control Plane Security** section. This section includes three toggle switches: **Enable CPsec:** (5), **Enable auto cert provisioning:** (5), and **Only accept APs from specified ranges:**. A link for **Cluster for Allowlist Propagation** is also visible.

D. Configure VLANs

- Create VLANs
- Configure VLAN Interface
- Assign VLANs on port

Create VLANs

The screenshot displays the Aruba Mobility Controller web interface. At the top, the Aruba logo is on the left, followed by the system name 'ArubaMC_Group99'. To the right, there are status indicators for 'ACCESS POINTS' (0), 'CLIENTS' (0), and 'ALERTS' (0). A user dropdown menu shows 'admin'. The left sidebar contains a menu with 'Configuration' and 'Interfaces' highlighted. The main content area shows the 'VLANs' tab selected, with a table containing one entry (ID 1) and a '+ 5' button to add more.

MOBILITY CONTROLLER
ArubaMC_Group99

ACCESS POINTS 0 0
CLIENTS 0 0 0
ALERTS 0

Mobility Controller

Dashboard
Configuration
WLANs
Roles & Policies
Access Points
AP Groups
Authentication
Services
Interfaces
System
Tasks
License
Redundancy

Ports **VLANs** IP Routes GRE Tunnels Pool Management OSPF Multicast

VLANs

NAME	ID(S)
--	1

+ 5

Configure VLAN Interface

Mobility Controller > **ArubaMC_Group99**

Dashboard

Configuration

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces**
- System
- Tasks
- Redundancy
- IoT

Diagnostics

Maintenance

Ports **VLANs** Routes IPv6 Neighbors GRE Tunnels Pool Management

VLANs

NAME	ID(S)	
Client	10,20,30	
--	1,98	

VLANs > Client

ID	IPV4 AD...	IPV6 AD...	ENABLE ...	PORT M...	ADMIN ...	OPERATI...	PD CLIE...	DHCP SE...
10	--	--	--	0/0/0	--	--	Disabled	None
20	--	--	--	0/0/0	--	--	Disabled	None
30	--	--	--	0/0/0	--	--	Disabled	None

Assign VLANs on port

Mobility Controller > ArubaMC_Group99

Dashboard
Configuration
WLANs
Roles & Policies
Access Points
AP Groups
Authentication
Services
Interfaces
System
Tasks
Redundancy
IoT
Diagnostics
Maintenance

Ports VLANs IP Routes IPv6 Neighbors GRE Tunnels Pool Management

Port Channel

NAME	MEMBERS	PROTOCOL	TRUSTED	POLICY	MODE	NATIVE VLAN	TRUNK VLANS
+							

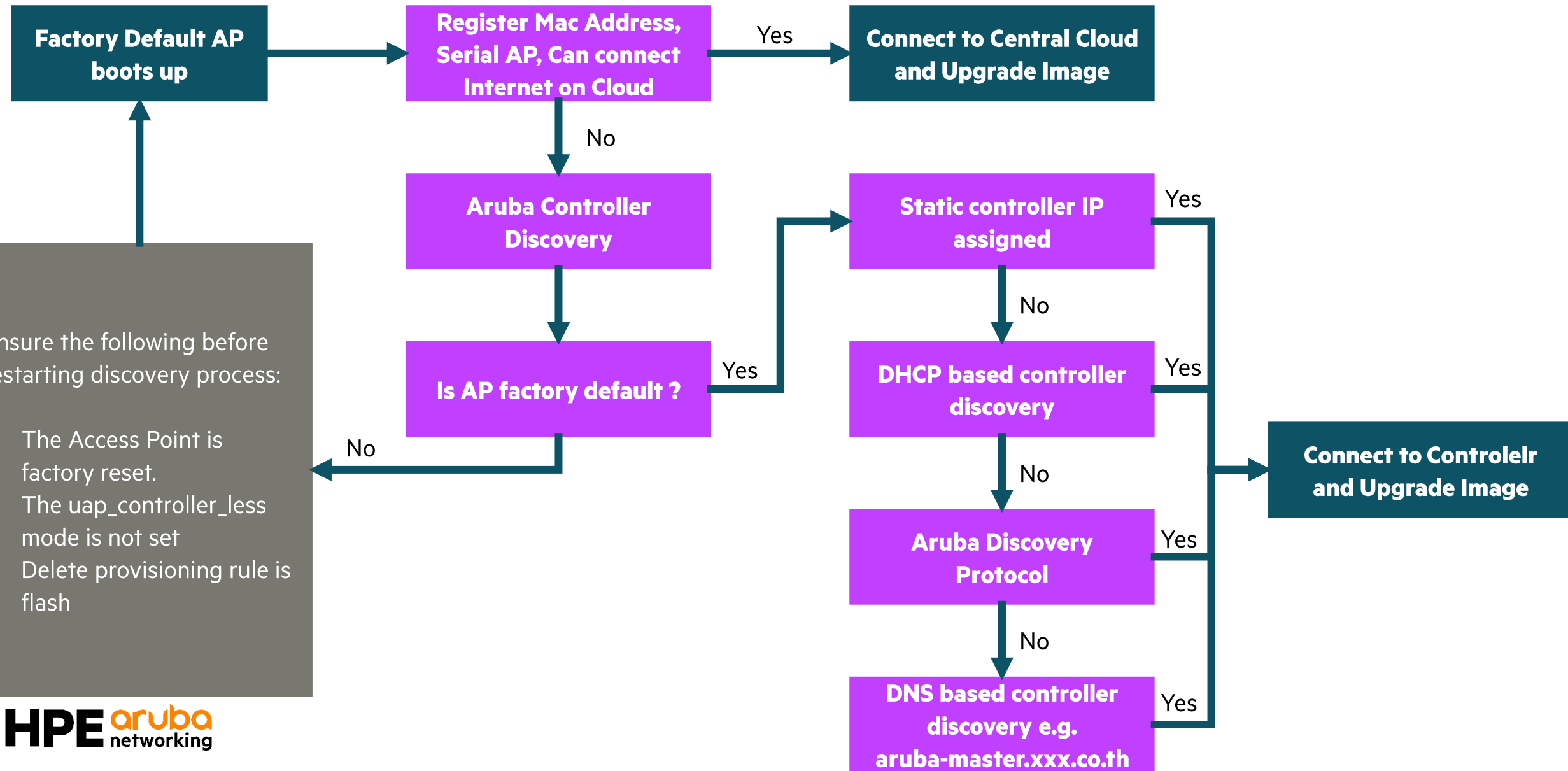
Ports

PORT	ADMI...	TRUST...	POLICY	MODE	NATIV...	ACCES...	TRUN...	SPAN...	MONI...	DESC...
GE-0/...	Enabl...	✓	Not-d...	trunk	98	1	1-4094	✓	--	--
GE-0/...	Enabled	✓	Not-d...	access	1	1	1-4094	✓	--	--
GE-0/...	Enabled	✓	Not-d...	access	1	1	1-4094	✓	--	--
GE-0/...	Enabled	✓	Not-d...	access	1	1	1-4094	✓	--	--

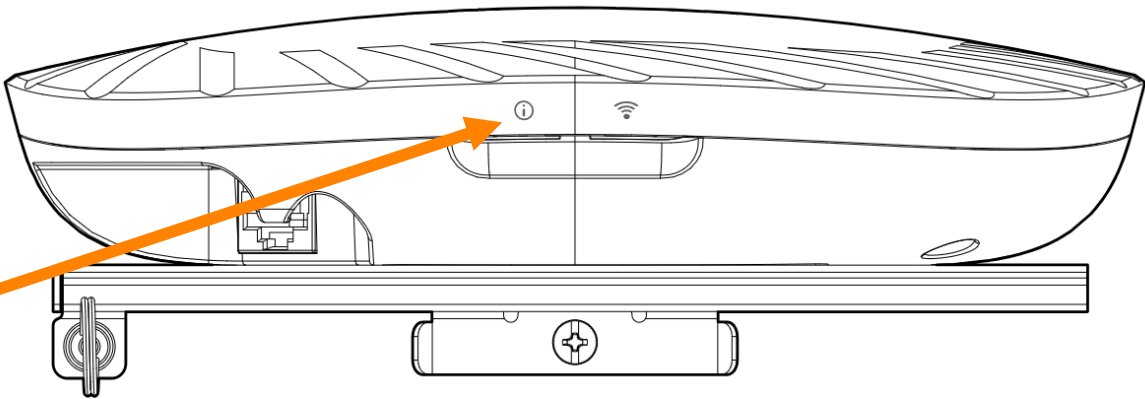
E.Initial Access Point(AP)

- **Understand Flow Unified**
- **LED Status**
- **Reset Setup**
- **AP convert to Controller**

Understand Flow Unified AP



LED Status

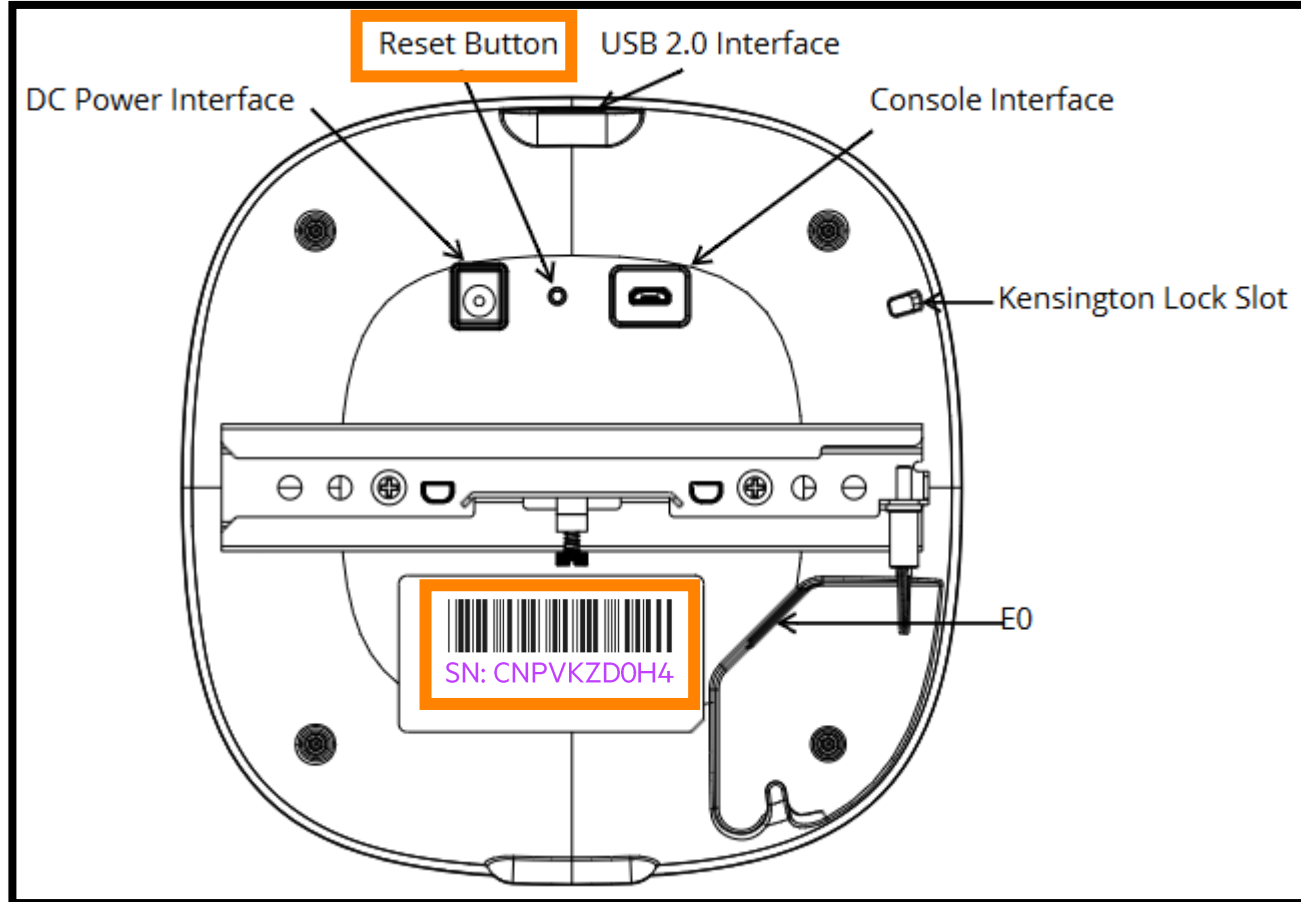


System Status LED ⓘ

Color/State	Meaning
Off	Device Powered off
Green - solid	Device ready, fully functional, no network restrictions
Green - blinking ¹	Device booting, not ready
Green - flash on ²	Device ready, fully functional, uplink negotiated in sub-optimal speed (<1Gbps)
Green - flash off ³	Device in deep-sleep mode
Amber - solid	Device ready, restricted power mode (IPM restrictions applied), no network restrictions
Amber - flashing off	Device ready, restricted power mode (IPM restrictions applied), uplink negotiated in sub-optimal speed (<1Gbps)
Red	System error conditions – Immediate action required

1. Blinking: one second on, one second off, 2 seconds cycle.
2. Flashing off: mostly on, fraction of a second off, 2 seconds cycle.
3. Flashing on: mostly off, fraction of a second on, 2 seconds cycle.

Reset Setup



Opt-1 Reset the AP during normal operation

press and hold down the reset button using a small, narrow object such as a paper clip for more than 10 seconds during normal operation.

Opt-2 Reset the AP while powering up

1. Press and hold down the reset button using a small, narrow object such as a paper clip while the access point is not powered on (either via DC power or PoE).
2. Connect the power supply (DC or PoE) to the access point while the reset button is being held down.
3. Release the reset button on the access point after 15 seconds.

Next Step

Access Point broadcast SSID : SetMeUp-XX:XX:XX

 SetMeUp-C2:69:D4

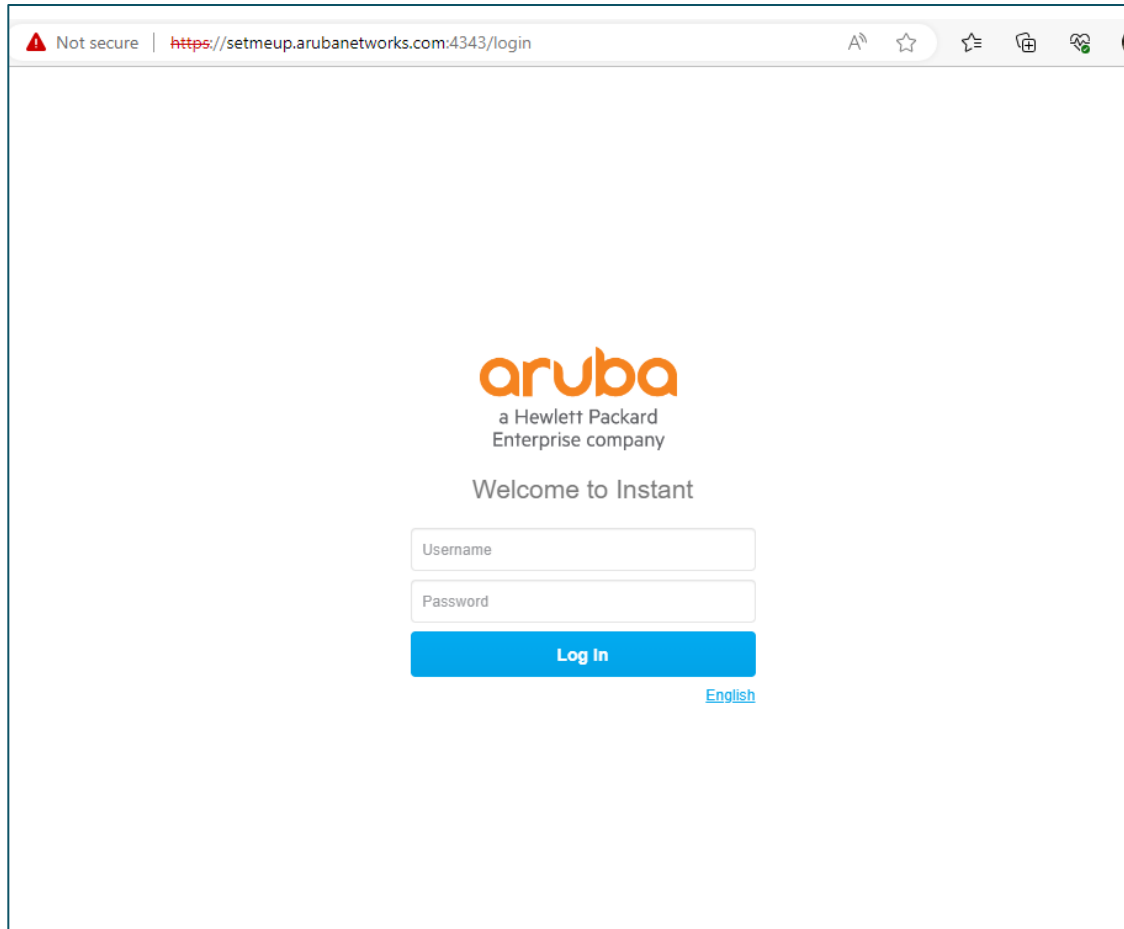
Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)

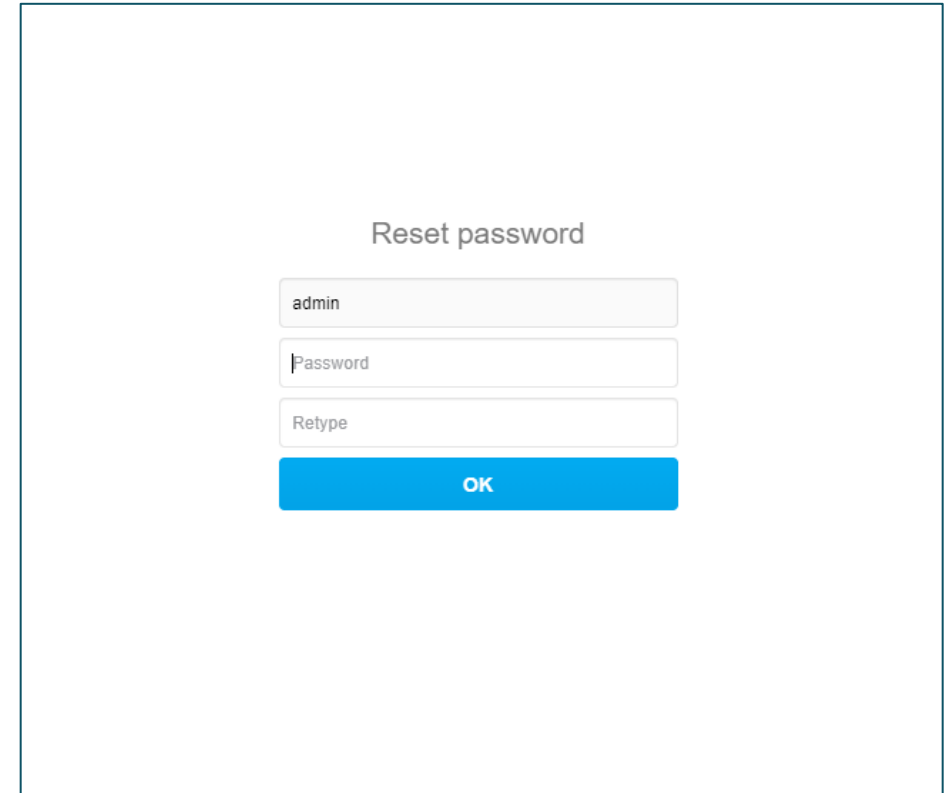
Login GUI and Changes password

Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)



A screenshot of a web browser showing the Aruba Instant login page. The browser's address bar displays "https://setmeup.arubanetworks.com:4343/login". The page features the Aruba logo (a Hewlett Packard Enterprise company) and the text "Welcome to Instant". Below this, there are two input fields labeled "Username" and "Password", followed by a blue "Log In" button. A link for "English" is visible at the bottom right of the login area.



A screenshot of the "Reset password" page. It contains three input fields: the first is pre-filled with "admin", the second is labeled "Password", and the third is labeled "Retype". Below these fields is a blue "OK" button.

Enter new **Password: P@ssw0rd** and **Retype password: P@ssw0rd**

Select Country Code

Select : **TH-Thailand**

Overview

Info

Networks

Access Points

Clients

Name

SetMeUp-C2:69:D4

Country code

Management

Uplink type

Welcome to Instant

Please specify the Country Code

TH - Thailand

1

RO - Romania

RS - Serbia

RU - Russia

RW - Rwanda

SA - Saudi Arabia

SE - Sweden

SG - Singapore

SH - Saint Helena

SI - Slovenia

SJ - Svalbard and Jan Mayen

SK - Slovak Republic

SM - San Marino

SN - Senegal

SR - Suriname

SS - South Sudan

SV - El Salvador

SX - Sint Maarten

TC - Turks And Caicos Islands

TF - French Southern Territories

TH - Thailand

please try [Upgrading your AP](#) to the latest version and try again, please contact technical support for more information

Dashboards-> Overview

aruba

VIRTUAL CONTROLLER

SetMeUp-C2:69:D4

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Alerts

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Active

Inactive

Up

Down

Wireless

Wired

1

0

1

0

1

0

Dashboard

IDS

AirGroup

0 Alerts

Clients

Throughput (bps)

2

1

0

12:28

12:29

12:30

12:31

12:32

1M

10K

100

0

100

10K

1M

12:28

12:29

12:30

12:31

12:32

Out

In

RF Dashboard

Clients

Signal

Speed

Access Points

Utilization

Noise

Errors

All Clients

9c:8c:d8:c2:6...

Convert Access Point Campus mode

aruba | VIRTUAL CONTROLLER | SetMeUp-C2:69:D4

Dashboard

- Overview
- Networks
- Access Points
- Clients
- Mesh Devices

Configuration

Maintenance 1

- About
- Firmware
- Configuration
- Certificates
- Reboot

Convert

Convert one or more Access Points to

Hostname or IP Address of Mobility Controller

After conversion, all Access Points will be managed by the Controller specified above.

Convert

Campus APs managed by a Mobility Controller 1

- Remote APs managed by a Mobility Controller
- Campus APs managed by a Mobility Controller
- Standalone AP
- Single AP

aruba | VIRTUAL CONTROLLER | SetMeUp-C2:69:D4

Dashboard

- Overview
- Networks
- Access Points
- Clients
- Mesh Devices

Configuration

Maintenance 1

- About

Convert

Convert one or more Access Points to

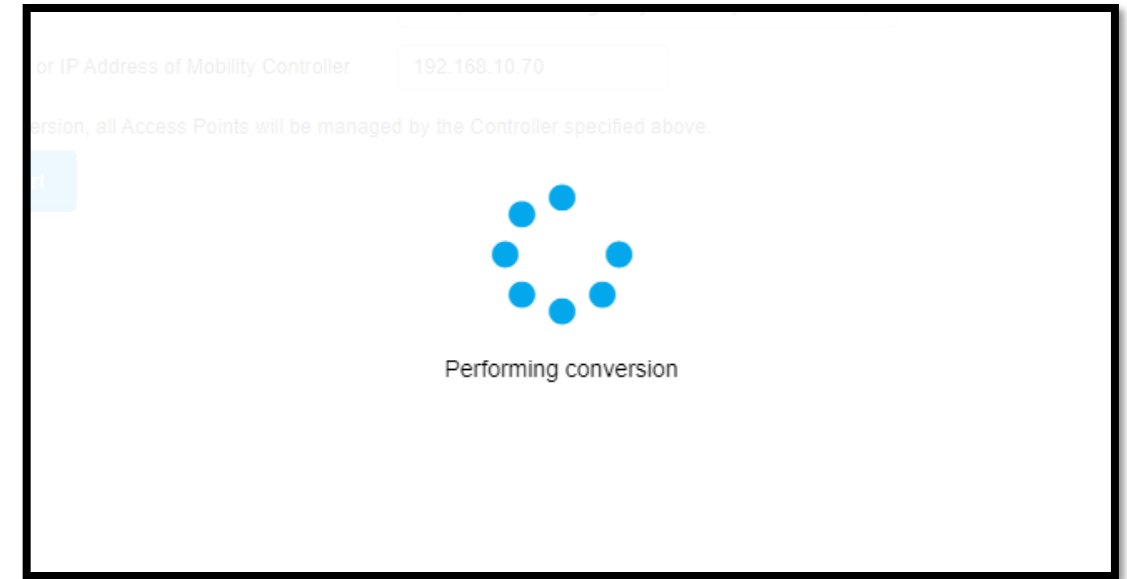
Hostname or IP Address of Mobility Controller

172.16.98.1-7

After conversion, all Access Points will be managed by the Controller specified above.

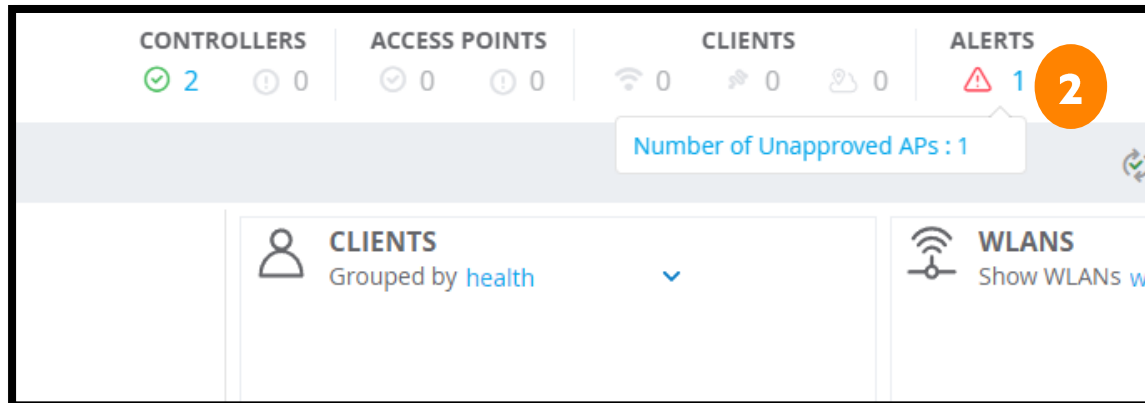
Convert 1

Convert Access Point Campus mode (Cont.)



Approve device on Mobility Controller(MC)

In-Case: enable auto cert provisioning not yet.



F. Create AP Group

- Provision AP Change to Group

Create Group AP

aruba

MOBILITY
CONTROLLER
ArubaMC_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

🔊 0

👤 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller

Dashboard

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Roles & Policies

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License

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AP Groups 2

NAME	APs
default	1
NoAuthApGroup	--

+

New AP Group

Name:

Group-99

Cancel

Submit

HPE

aruba
networking

Change AP to New Group

Mobility Controller > 1

Dashboard

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WLANs

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Campus APs Remote APs Mesh APs Allowlist Provisioning Rules

Campus APs 1

☒	AP NAME	AP GROUP	IPV4 ADD...	IPV6 ADD...	SWITCH IP	MAC ADD...	SERIAL #	T
☒	7c:57:3c:c...	default	192.168.9...	--	192.168.9...	7c:57:3c:c...	CNPBK51...	3

5 Provision

Flags:

U = Unprovisioned, N = Duplicate name, G = No such group, L = Unice...

7c:57:3c:c3:e0:e4

MAC address: 7c:57:3c:c3:e0:e4

LLDP neighbor chassis ID / port ID: 6200 / 1/1/3

Name: 7c:57:3c:c3:e0:e4

AP group: Group-99 6 Select New Group

Controller discovery: ☒ Use AP discovery protocol (ADP) ☐ Static

Controller discovery preference: ☒ IPv4 ☐ IPv6

IP: ☒ DHCP ☐ Static

Deployment: ☒ Campus ☐ Remote ☐ Mesh ☐ Remote mesh portal

Wi-Fi uplink: ☐ Wi-Fi uplink

Cancel Submit 6

Access Points will be Rebooted

CAUTION: Applying this configuration change will interrupt service while the affected Access Points are rebooted.

Do you want to continue ?

Cancel Continue & Reboot 7

A. Configure WLAN PSK+MAC

- Add local user

Configure WLAN PSK

aruba

MOBILITY
CONTROLLER
ArubaMC_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠️ 0

?

admin ▾

Mobility Controller 1

Dashboard

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Roles & Policies

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WLANs 0

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION	
+ 3				

Configure WLAN PSK

WLANs 1

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION	
PSK-G99	default, Group-99	WPA2-Personal	--	

PSK-G99

GeneralVLANsSecurityAccessProfiles

Name (ssid):

PSK-G99

1 Create WiFi Name

Primary usage:

☒ Employee ☐ Guest

Select AP Groups

Broadcast on:

☐ default ☒ Group-99

2 Select New GroupAP

Forwarding mode:

Tunnel

Cancel

Submit

3

Configure WLAN PSK

aruba

MOBILITY CONTROLLER
ArubaMC_Group99

ACCESS POINTS

1

0

CLIENTS

1

0

0

ALERTS

0

admin

Mobility Controller

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Client

WLAN

265 kB

Radios

Wireless Clients 1

NAME	IP ADDRESS	HEALTH	BAND	ROLE	SNR	USAGE	WLAN	CONNECTED...
192.168.20.1	192.168.20.1	Good	5 GHz	authenticated	54 dB	104 kB	PSK-G99	7c:57:3c:c3:e0:e4

DETAILS

Name
192.168.20.1

IP address
192.168.20.1

MAC address
a4:d9:31:b8:57:75

Health score
92%

Speed
842 Mbps

Max speed
866 Mbps

Frames in the last minute
121

SIGNAL

Show information about signal quality

100

75

50

25

0

20:21

20:22

Now

USAGE

Show information about throughput

10k

0

10k

20k

20:22

Now

Transmitted

Received

Copy mac address

Add mac address to local user

aruba

MOBILITY CONTROLLER
ArubaMC_Group99

ACCESS POINTS
1 0

CLIENTS
1 0 0

ALERTS
0

admin

Mobility Controller

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Auth Servers3

Profiles

L2 Authentication

L3 Authentication

User Rules

Advanced

Server Groups 2

NAME	SERVICES	FAIL THROUGH	LOAD BALANCE	SERVICES RULES	
default	1	--	--	1	
internal	1	--	--	1	4

+

Server Group > internal

ServersOptionsServer Rules

Drag rows to re-order

NAME	TYPE	IP ADDRESS	TRIM FQDN	MATCH RULES	
Internal	--	--	--	0	5

+

Server Group > internal > Internal

UsersOptionsImportExportServer Group Trim FQDNServer Group Match RulesGuest user page

USER NAME	ROLE	EMAIL	EXPIRATION	STATIC IP FOR RAPS	ENABLED	
-----------	------	-------	------------	--------------------	---------	--

+

6

Add mac address to local user

1 Past mac address to username, password, retype password

Internal Server > Add New User

User name:	a4d931b85775	Generate
Password:		Generate
Retype password:		
Role:	authenticated	2
E-mail:		
Static inner IP for RAPs:		
Enabled:	☒	
Expiration:	None	

Cancel Submit 3

Configure WLAN Add MAC Authentication

1. The **Mobility Controller** menu item is highlighted in the top navigation bar.

2. The **WLANs** option is selected in the left sidebar under the **Configuration** section.

3. The **WLANs** table is displayed with the following data:

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
PSK-G99	default	WPA2-Personal	--

A blue plus icon is visible below the table to add a new WLAN.

The **Security** tab is selected for the **PSK-G99** WLAN configuration.

On the left, a security slider is positioned at **Personal**, between **Enterprise** and **Open**.

On the right, the following settings are visible:

- Key management: WPA3-Personal (dropdown)
- ☒ Enable backward compatibility
- Passphrase: [Redacted]
- Retype: [Redacted]
- MAC authentication: Enabled (dropdown)**
- Denylisting: [Off]

4. The **MAC authentication** dropdown is highlighted, indicating it should be set to **Enabled**.

Text annotation: **Enable MAC authentications**

Buttons: **Cancel** and **Submit** are located at the bottom right.

Configure WLAN Add MAC Authentication

PSK-G99 General VLANs Security **Access** Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

Cancel **Submit**

Pending Changes

WLANs 1

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
PSK-G99	default	WPA2-Personal	--

General **VLANs** **Security** **Access** Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

Pending Changes

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

Close **Discard changes** **Deploy changes**

Configure WLAN Add MAC Authentication

PSK-G99 General VLANs Security **Access** Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

Cancel **Submit**

Pending Changes

WLANs 1

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
PSK-G99	default	WPA2-Personal	--

General **VLANs** **Security** **Access** Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

Pending Changes

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

Close **Discard changes** **Deploy changes**

B. Configure WLAN 802.1X

Configure WLAN 802.1X

aruba

MOBILITY
CONTROLLER
ArubaMC_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠️ 0

?

admin ▾

Mobility Controller

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Tasks

WLANs 0

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION	
+ 3				

Configure WLAN 802.1X

New WLAN

General VLANs Security Access

Name (SSID): 1

Primary usage: ☒ Employee ☐ Guest

Select AP Groups

Broadcast on: ☐ default ☒ Group-99 2

Forwarding mode: Tunnel

3

New WLAN

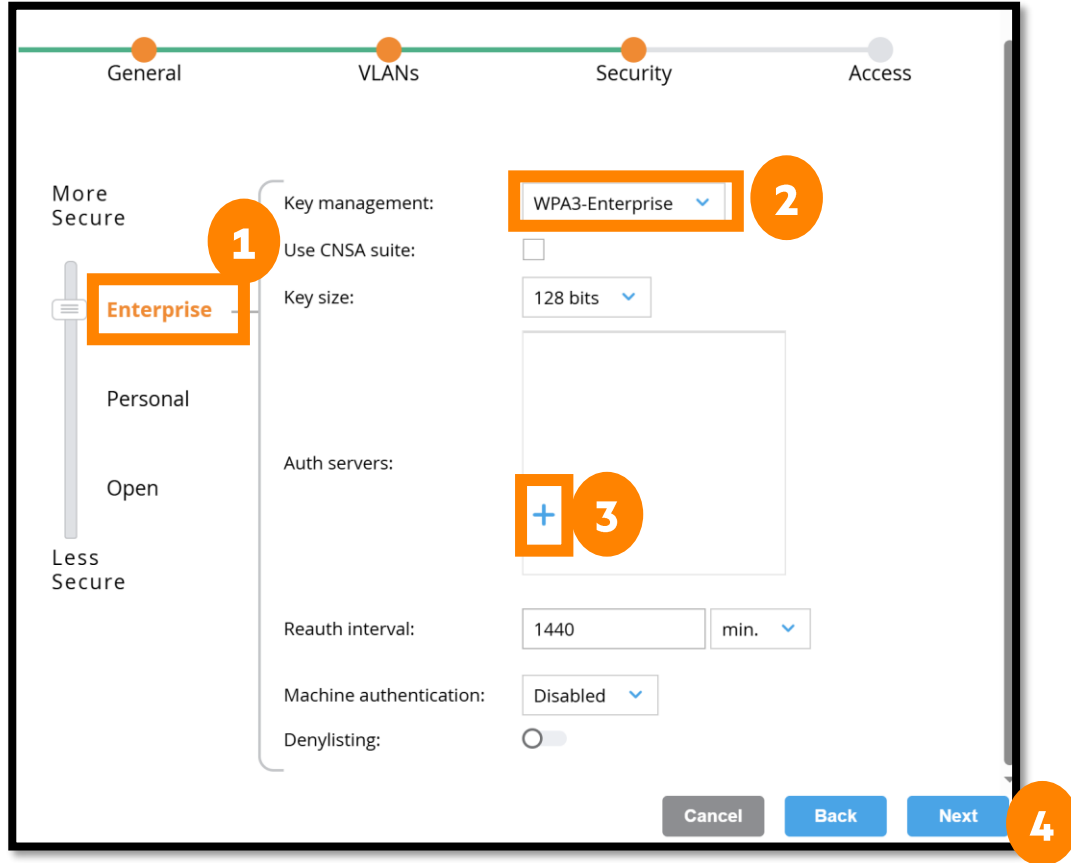
General VLANs Security Access

VLAN: 10 4 Select vlan 10

[Show VLAN details](#)

5

Configure WLAN 802.1X



General VLANs Security Access

More Secure

Enterprise

Personal

Open

Less Secure

Key management: WPA3-Enterprise

Use CNSA suite: ☐

Key size: 128 bits

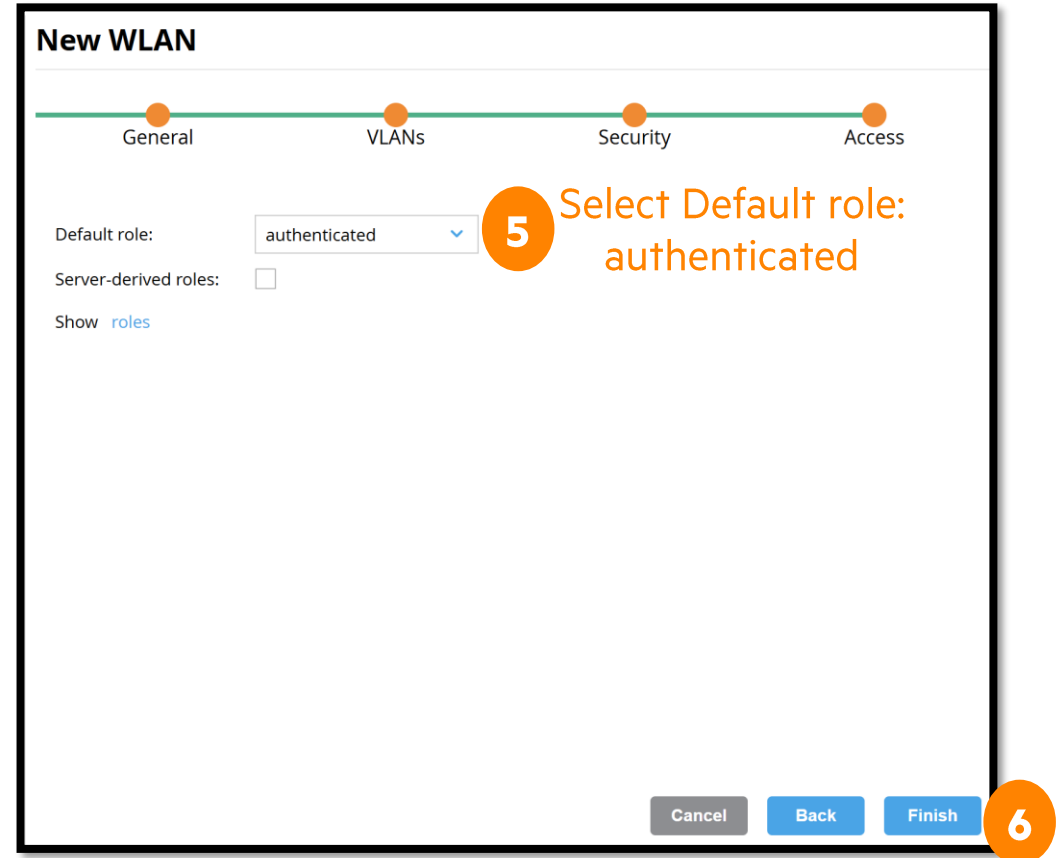
Auth servers: +

Reauth interval: 1440 min.

Machine authentication: Disabled

Denylisting: ☐

Cancel Back Next



New WLAN

General VLANs Security Access

Default role: authenticated

Server-derived roles: ☐

Show roles

Cancel Back Finish

Select Default role: authenticated

Configure add local user for 802.1X

Mobility Controller

Dashboard **Configuration** (1) Roles & Policies Access Points AP Groups **Authentication** (2) Services Interfaces System Tasks License Redundancy Diagnostics Maintenance

Auth Servers (3) AAA Profiles L2 Authentication L3 Authentication

Server Groups 3

NAME	SERVICES	FAIL THROUGH	LOAD BALANCE	SERVER RULES
default	1	--	--	1
Internal	1	--	--	1
802.1X-G99_dot...	1	--	--	0

All Servers 1

NAME	TYPE	IP ADDRESS / HOST	SERVER GROUP
Internal	--	--	default internal 802....

Auth Servers AAA Profiles L2 Authentication L3 Authentication

Server > Internal **Users** Options Import Export [Guest user page](#)

USER NAME	ROLE	EMAIL	EXPIRATION	STATIC IP FO...	ENABLED
a4d931b857...	authenticated	--	--	0.0.0.0	Yes

Internal Server > Add New User (6)

User name: group99 **Generate**

Password: **Generate**

Retype password:

Role: authenticated (7)

E-mail:

Static inner IP for RAPs:

Enabled: ☒

Expiration: None

Create User
User: group0X
Pass: P@sswOrd

Submit (8) Cancel

C. Configure WLAN Guest

- Add local user

Configure WLAN Guest

aruba

MOBILITY
CONTROLLER
ArubaMC_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller

1

Dashboard

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2

WLANs

Roles & Policies

Access Points

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Tasks

WLANs 0

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION	
+ 3				

Configure WLAN Guest

New WLAN

General VLANs Security Access

Name (SSID): 1

Primary usage: ☐ Employee ☒ Guest 2

Select AP Groups

Broadcast on: ☐ default ☒ Group-99 3

Forwarding mode:

Cancel Back Next 4

New WLAN

General VLANs Security Access

VLAN: 5 Select vlan 30

[Show VLAN details](#)

Cancel Back Next 6

D. Configure Role & Policy

- **Network aliases**

Network Aliases

Configure on node “Mobility Controller”-> Configuration-> Roles & Policies -> Aliases -> +

The screenshot displays the Aruba Mobility Conductor web interface. At the top, the header includes the Aruba logo, the system name "MOBILITY CONDUCTOR ArubaOS_MM01_Ha...", and status indicators for Controllers (2 green, 0 yellow), Access Points (0 green, 0 yellow), Clients (1 Wi-Fi, 0 Bluetooth, 0 NFC), and Alerts (0). A user menu shows "admin" with a dropdown arrow. The left sidebar contains a navigation menu with "Dashboard" and "Configuration" expanded, showing sub-items like WLANs, Roles & Policies (highlighted), Access Points, AP Groups, Authentication, Services, Interfaces, Controllers, System, and Tasks. The main content area has tabs for "Roles", "Policies", "Applications", and "Aliases" (selected). Under the "Aliases" tab, there is a section for "Network Aliases" containing a table with the following data:

NAME	ITEMS	DESCRIPTION	IP VERSION	INVERT	
mswitch	1	--	IPv4	--	
controller	1	--	IPv4	--	
vrrp_ip	1	--	IPv4	--	
any	1	--	IPv4	--	
user	1	--	IPv4	--	
localip	1	--	IPv4	--	
auth-google	2	--	IPv4	--	

Below the table is a blue "+" icon to add a new alias. A small grid icon is visible in the top right corner of the table area.

Network Aliases (cont.)

Enter Name "Local_Network" +

Destination Local_Network

IP version: 1

Name:

Description:

Invert: ☐

Items

TYPE	IP ADDRESS	NETMASK/RANGE/...	DOMAIN NAME
------	------------	-------------------	-------------

+ 2

Rule type : **Network**
IP Address : **192.168.0.0**
Network mask : **255.255.0.0**
OK -> Submit

Add New Destination

Rule type: 3

IP address:

Network mask:

4

Policies

Create Policies

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Roles

Policies

Applications

Aliases

NAME	RULES COU...	TYPE	POLICY US...	DESCRIPTION
validuseret...	1	eth	--	--
global-sacl	0	session	guest, statef...	--
sdn-acl	0	session	--	--
sys-control	13	session	sys-ap-role, ...	--
sys-ap-acl	12	session	sys-ap-role	--
switch-logo...	1	session	switch-logon	--
sys-switch-acl	12	session	sys-switch-r...	--

New Policy

Policy type:

Session

Policy name:

Local_Network-Polic

Description:

Cancel

Submit

Policies (cont.)

Roles **Policies** Applications Aliases

Policies

NAME	RULES C...	TYPE	POLICY ...	DESCRIP...	
apprf-sys-ap-role-sacl	0	session	sys-ap-role	--	
apprf-sys-switch-role-sacl	0	session	sys-switc...	--	
apprf-guest-99-guest-log...	0	session	Guest-99...	--	
Local_Network-Policies	0	session	--	--	1
uplink-lb-cfg-racl	0	routing	--	--	
uplink-lb-sys-racl	0	routing	--	--	
master-boc-traffic	0	routing	--	--	

Policy > Local_Network-Policies Rules ⓘ Drag rows to re-order

IP VERSI...	SOURCE	DESTINA...	SERVICE/...	ACTION	DESCRIP...	
-------------	--------	------------	-------------	--------	------------	--

2

New Rule for Local_Network-Policies

Rule type: ☒ Access control **3** ☐ Application

Cancel

OK **4**

Local_Network-Policies > New forwarding Rule

IP version: IPv4 **5**
Source: Any
Destination: Alias
Destination alias: local_network
Service/app: Service
Services alias: svc-icmp
Action: Deny

TOS:
Time range: - None - Reset
802.1p priority:
Options:
☐ Log ☐ Mirror ☐ Denylist ☐ Disable scanning
Queue:
Description:

Cancel

Submit **6**

Create Policies Block ping

IP Version : **IPV4**

Source : **Any**

Destination : **Alias**

Destination alias : **Local_Network**

Service/app : **Service**

Service alias : **svc-icmp**

Action : **Deny**

Submit

Role

Create Role

Mobility Controller

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License

Roles

Policies

Applications

Aliases

Roles 15

NAME	RULES
logon	32 Rules
guest	11 Rules
ap-role	35 Rules
stateful-dot1x	0 Rules
guest-logon	27 Rules
sys-ap-role	25 Rules
sys-switch-role	25 Rules

+



New Role

Name:

engineer

Cancel

Submit

Role

Assign policies in Role engineer

Mobility Controller >

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Roles

ies

Applications

Aliases

Roles 16

NAME	RULES
denyall	1 Rules
default-via-role	3 Rules
default-vpn-role	4 Rules
authenticated	4 Rules
voice	45 Rules
Guest-99-guest-logon	26 Rules
engineer	0 Rules

+

engineer

Show Advanced View

Global Rules

IP VERSI...	SOURCE	DESTINA...	SERVICE/...	ACTION	DESCRIP...
-------------	--------	------------	-------------	--------	------------

+

engineer

Policies

Bandwidth

Captive Portal

Show Basic View

NAME	RULES COU...	TYPE	POLICY US...	DESCRIPTI...
global-sacl	0	session	guest, state...	--
apprf-engin...	0	session	engineer	--
engineer	0	session	engineer	--

+

Role

Assign policies > local_network(first), allowall(second)

New Policy

☒ Add an existing policy **1** ☐ Create a new policy

Policy type: Session

Policy name: Local_Network-Policies **2**

Description:

Position:

3 Cancel Submit

first
→

engineer	Policies	Bandwidth	Captive Portal	Show Basic View
NAME	RULES COU...	TYPE	POLICY US...	DESCRIPTI...
global-sacl	0	session	guest, state...	--
apprf-engin...	0	session	engineer	--
engineer	0	session	engineer	--
Local_Netw...	1	session	--	--

4

5

second
↓

New Policy

☒ Add an existing policy **6** ☐ Create a new policy

Policy type: Session

Policy name: allowall **7**

Description:

Position:

8 Cancel Submit

save
→

Pending Changes **10**

Roles Policies Applications Aliases

Roles 16

NAME	RULES
denyall	1 Rules
default-via-role	3 Rules
default-vpn-role	4 Rules
authenticated	4 Rules
voice	45 Rules
Guest-99-guest-logon	26 Rules
engineer	3 Rules

9

engineer	Policies	Bandwidth	Captive Portal	Show Basic View
NAME	RULES COU...	TYPE	POLICY US...	DESCRIPTI...
apprf-engin...	0	session	engineer	--
engineer	0	session	engineer	--
Local_Netw...	1	session	engineer	--
allowall	2	session	default-iap...	--

Role

Assign role engineer to user local

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Auth Servers

AAA Profiles

L2 Authentication

Server Groups

NAME	SERVERS	FAIL THRO...	LOAD BAL...	SERVER RU...
default	1	--	--	1
internal	1	--	--	1
802.1X-G99...	1	--	--	0

All Servers 1

NAME	TYPE	IP ADDRESS / ...	SERVER GROUP
Internal	--	--	default intern...

Server > Internal

Users

Options

Import

Guest user page

USER NA...	ROLE	EMAIL	EXPIRATI...	STATIC IP...	ENABLED
a4d931h	authenti...	--	--	0.0.0.0	Yes
group99	authenti...	--	--	0.0.0.0	Yes
guest99	guest	--	--	0.0.0.0	Yes



Edit User group99

User name:group99

Password:*****Generate

Role:engineer

E-mail:

Static inner IP for RAPS:0.0.0.0

Enabled:☒

Expiration:None

CancelSubmit

Role

Test connect WiFi > 802.1X-G99

ACCESS POINTS

1

0

CLIENTS

1

0

0

ALERTS

0

admin

Search

1

3

82.1 kB

2

Wireless Clients 1

NAME	IP ADDR...	HEALTH	BAND	ROLE
group99	192.168.10.2	Unkno...	5 GHz	engineer

Test ping gateway user

Command Prompt

Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sakkr>ping 192.168.10.254

Pinging 192.168.10.254 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.254:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\sakkr>ping 192.168.10.253

Pinging 192.168.10.253 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.253:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

CUSTOMER FIRST
CUSTOMER LAST

The image features the text "CUSTOMER FIRST" on the top line and "CUSTOMER LAST" on the bottom line. The letters are large, bold, and white. Each letter is filled with a different photograph of people at what appears to be a conference or event. The photos show various individuals, some smiling, some looking at devices, and others in group settings. The overall theme is customer-centricity.